



Global public health intelligence report 2024



World Health
Organization

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Foreword

In an increasingly interconnected and fragile world, the early detection, risk assessment and rapid response to public health threats are more critical than ever. Recent years have demonstrated how quickly emerging and re-emerging health threats can disrupt societies, strain economies and test the resilience of health systems. At the same time, protracted conflicts, large-scale displacement and widening gaps in humanitarian financing continue to challenge collective efforts to protect the health and dignity of millions of people worldwide.



Remarkable progress has been made in building national capacities and strengthening international collaboration, however, the frequency and complexity of public health emergencies remind us that robust and timely public health intelligence remains a cornerstone of effective preparedness and response, as outlined in the *WHO Thirteenth General Programme of Work* (GPW 13). From infectious disease outbreaks to disasters, chemical incidents and the health impacts of conflict, World Health Organization's (WHO) Public Health Intelligence work underpins rapid response and informs decisions that can prevent an event from escalating into a wider crisis. This fourth edition of the *Global public health intelligence report 2024* showcases how WHO and its partners continue to deliver on this commitment under the International Health Regulations, even in highly complex contexts and despite severe resource constraints.

The report presents key findings and trends from 2024, including progress in strengthening the timeliness of detection, verification and information sharing in line with the 7–1–7 target — to detect a public health threat within 7 days, notify within 1 day and complete an initial risk assessment and response within 7 days. Notably, advances in integrated surveillance systems, the expansion of the Epidemic Intelligence from Open Sources (EIOS) platform and the application of One Health approaches have played a vital role in enhancing early warning capabilities at the human–animal–environment interface.

Throughout 2024, WHO Public Health Intelligence teams across WHO headquarters, regional and country offices have worked tirelessly with Member States, partners and communities to identify, risk assess and communicate critical information on hundreds of acute public health events. These efforts have ensured that timely and accurate data support local, national and global actions, helping to protect lives in both routine and highly challenging settings, including fragile, conflict-affected and vulnerable contexts.

In the current challenging scenario, marked by critical funding shortages and escalating conflicts across different regions, WHO will continue to spearhead this critical area of work and make every effort to enhance public health intelligence operations, advocate for best practices and share timely, actionable evidence for the benefit of individuals and communities worldwide — to protect lives, reduce suffering and ensure that no one is left behind.



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The *Global public health intelligence report 2024* represents the fourth edition of the report and builds on the three previous *Global public health intelligence reports*, as well as previous bi- and tri-regional public health intelligence reports produced by the World Health Organization (WHO) regional offices for the Americas, Europe and Africa.

This report is a collective effort from all Public Health Intelligence (PHI) teams across the WHO regional offices and the headquarters Department of Health Emergency Alert, Response and Operations (EOP) within the WHO Health Emergencies Programme (WHE).

A core team coordinated the production of the report, including Lidia Alexandrova Ezerska, Yosef Asrat Temre, Adedoyin Awofisayo-Okuyelu, Esther Hamblion, Alessandro Miglietta and the Health Information Management team in the WHO Regional Office for Europe led by Silviu Ciobanu.

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Abbreviations

COVID-19	coronavirus disease 2019
cVDPV2	circulating vaccine-derived poliovirus type 2
DON	Disease Outbreak News
EBS	event-based surveillance
EC	European Commission
ECDC	European Centre for Disease Prevention and Control
EFSA	European Food Safety Authority
EIS	WHO Event Information Site
EIOS	Epidemic Intelligence from Open Sources
EMS	Event Management System
ERF	Emergency Response Framework
EU/EEA	European Union/European Economic Area
FAO	Food and Agriculture Organization of the United Nations
GCDC	Gulf Center for Disease Prevention and Control
GDACS	Global Disaster Alert and Coordination System
GLEWS+	Joint FAO–WHO–WOAH Global Early Warning System
GOARN	Global Outbreak, Alert and Response Network
ICC	International Cricket Council
IHR	International Health Regulations
MERS	Middle East respiratory syndrome coronavirus
NFP	National IHR Focal Point
PHEIC	Public Health Emergency of International Concern
PHSA	Public Health Situation Analysis
RKI	Robert Koch Institute
RRA	Rapid Risk Assessment
UNEP	United Nations Environment Programme
WHO	World Health Organization
WOAH	World Organisation for Animal Health

Glossary

DETECTION

Detection is a process by which events of potential international public health concern are identified and recorded. WHO detects signals (see: *Signal*) using various sources. These include: (1) media through the Epidemic Intelligence from Open Sources (EIOS) system; (2) reporting through governmental channels, such as the Ministry of Health, national or sub-national government agencies and public health centres or institutes; (3) other United Nations agencies;

(4) partner networks, such as the Global Outbreak, Alert and Response Network (GOARN); and (5) the Joint FAO–WHO–WOAH Global Early Warning System (GLEWS+). Events are also directly reported by States Parties to WHO through National International Health Regulations (IHR) Focal Point (NFP) or comparable [European Commission (EC) Early Warning and Response System (EWRS) for EU/EEA Member States] channels.

DISEASE OUTBREAK NEWS REPORT

Disease Outbreak News (DON) reports provide verified information on acute public health events to the public, which are shared by WHO under the IHR. They provide validated epidemiological information on confirmed or potential acute

public health events that have significant public health implications or the potential to become a public health emergency of international concern (see: *PHEIC*).

EVENT

Event is defined by Article 1 of the IHR as “a manifestation of disease or an occurrence that creates a potential for disease”. This can include events that are infectious, zoonotic, food safety, chemical, radiological or nuclear in origin and transmitted by persons, vectors, animals, goods/food or through the environment. Due to inherent differences in pathogen presence, epidemiological dynamics, socio-political factors, climatic conditions and other disease drivers that might constitute an event in one

geographical area at a certain time, but might not be sufficient to meet the threshold for an event in another area, at a comparable or different time. Therefore, the categorization of events will vary between WHO regions. When comparing events between WHO regions, the underlying dynamics, like population vulnerability and capacity to respond – which influence the emergence, spread and impact of acute public health events – should always be taken into account.

EVENT DESIGNATION

Event designation is assigned during the risk assessment process to events recorded in the internal WHO Event Management System (EMS)

(see: *Event Management System*). There are six designation categories listed below.

1. *Substantiated*, when the presence of a hazard is confirmed, or the number of human cases exceeds established thresholds.
2. *Discarded*, when no international risk is expected or, after assessment, not deemed to be a significant public health risk.
3. *No outbreak*, when no human cases are reported (for example, in instances of outbreaks among animals), or when the number of human cases reported does not meet the threshold of an outbreak, or when the occurrence of the hazard is within expected levels.
4. *Unverifiable*, when no information is forthcoming from the NFP or responsible national authority to substantiate its occurrence, despite the best efforts to obtain such information.
5. *Under verification*, when the event continues to be undergoing the process of verification.
6. *No designation*, when no information regarding the designation is available.

EVENT INFORMATION SITE FOR IHR NATIONAL FOCAL POINTS

Event information site (EIS) is a secure website developed by WHO to facilitate communications with the IHR National Focal Points (NFPs) as part of the implementation of the IHR. Information on the site is provided by WHO to NFPs in confidence as specified in Article 11, paragraph one, of the IHR. Information sharing on the EIS occurs through two modalities:

1. **EIS Bulletins:** a publication on the EIS about a public health event occurring in a single State Party.
2. **EIS Announcement:** a publication on the EIS that can vary in nature and format. It may include, but is not limited to, temporary or standing recommendations issued under the IHR, information about an event occurring in multiple States Parties, WHO rapid alerts concerning medical products and other relevant updates.

EVENT MANAGEMENT SYSTEM

Event Management System (EMS) is the central electronic repository for event-related information. The EMS is a password-protected web-based tool accessible to designated personnel at WHO country and regional offices, and headquarters, subject to completion of relevant training. The key objectives of EMS are to support WHO users in the sharing and analysing of information on

events as well as reviewing the assessed risk to guide public health actions. There is a linkage between the EMS and the EIS to facilitate information sharing. The EMS is not intended to be an exhaustive database of all acute public health events occurring worldwide. Rather, its objective is to support the internal management of WHO for acute public health events.

HAZARD*

Hazard refers to the nature or type of the event. WHO has adopted an all-hazards approach, in which events of any nature, including infectious diseases, but also disasters, chemical, biological or nuclear threats and human-made conflicts, are assessed. Hazards have been grouped into seven categories, which include the following:

1. *Animal or zoonosis*, if there is potential harm to public health from an animal or zoonosis.
2. *Chemical*, if there is potential harm to public health from the toxic effects of chemical substances, which are chiefly non-medical, as to source.
3. *Disaster*, if there is potential harm to public health from a natural disaster.

* During 2022, selected hazard types were reclassified in the EMS. The following hazard classifications that were used in previous reports (animal, zoonosis and food safety) were reclassified to other respective hazards, and therefore, are no longer shown as separate classifications.

4. *Food safety*, if there is potential harm to public health from the toxic effects of food (poisoning or injury).
5. *Infectious*, if there is potential harm to public health from an infectious disease.
6. *Product*, if there is potential harm to public health from contaminated or faulty therapeutic goods including medicines, blood products, tissues and organs, medical devices, diagnostic tests and devices, etc. — including poisonings due to mislabelling of therapeutic goods.
7. Other, which includes the following hazards:
 - a. Nutritional deficiency, *if there is potential harm to public health from nutritional deficiencies*.
 - b. Radionuclear, *if there is potential harm to public health from the toxic effects of ionizing radiation*.
 - c. Societal, *if there are violent or hostile events, caused by humans, which have an impact on public health*.
 - d. Noncommunicable, *if there is potential harm to public health from a noncommunicable disease*.
 - e. Undetermined, *if there is potential harm to public health from an undetermined hazard*.

NATIONAL IHR FOCAL POINT

National IHR Focal Point is defined by Article 1 of the IHR as “the national centre, designated by each State Party, which shall be accessible at all times for communications with WHO

IHR Contact Points under these Regulations”. These are established as centres or offices rather than individuals.

PUBLIC HEALTH INTELLIGENCE

Public health intelligence (PHI) is defined by the United Nations terminology database as “a core public health function responsible for identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting and generating

a wide range of information for actionable insights and disseminating these for informed and effective decision-making to protect and improve the health of the population”.

PUBLIC HEALTH SITUATION ANALYSES

Specifically, the purpose of the Public Health Situation Analyses (PHSA) is to identify the current health status and potential health threats that the population may face, the functioning of the health system and the humanitarian health response, as defined in the *WHO Emergency Response Framework* (ERF 2.1). It is a review of the latest available secondary data. The PHSA aims to provide all health sector partners, including local and national authorities, nongovernmental

organizations, donor agencies and United Nations agencies with a common and comprehensive understanding of the public health situation in a crisis, in order to inform evidence-based collective humanitarian health response planning. The PHSA may also be used to feed other sectoral and intersectoral products, such as providing the health input to the Humanitarian Needs Overview, and is also used in support of the WHO (re-)grading process.

RAPID RISK ASSESSMENT

Rapid Risk Assessment (RRA) is a formalized WHO rapid risk assessment for acute events. It assesses the public health risk-based posed by a public health threat due to exposure to an identified

hazard. The RRA process involves a joint assessment by the WHO country and regional offices, and headquarters. It is conducted for events with serious public health implications following

pre-defined criteria within WHO. The RRA process provides a forum for the timely assessment of available data, which takes into account the contextual and hazard specific knowledge and feedback of key experts across WHO. It supports a collaborative expert prioritization of immediate actions in a time-sensitive manner. The finalized RRA report may be shared with key stakeholders that could contribute to the response.

RRA reports have become well accepted documents of high practical value both within WHO and among partners. The outcomes of the RRA are summarized in event bulletins published on the secure EIS for the NFPs and, when appropriate, on the public DON web page.

REQUEST FOR VERIFICATION

Request for verification is defined by Article 10 of the IHR as “WHO shall request, in accordance with Article 9, verification from a State Party of reports from sources other than notifications or consultations of events which may constitute a

public health emergency of international concern allegedly occurring in the State’s territory. In such cases, WHO shall inform the State Party concerned regarding the reports it is seeking to verify.”

RISK ASSESSMENT

Risk assessment is conducted at various stages of public health intelligence activities, including at the signal detection stage, when a signal is designated as an event, or an event is notified to WHO. For all events the risk assessment is documented at the stage when the event is first entered into

EMS against the following criteria: the potential public health impact, risk of international spread, risk of trade or travel restrictions and if the event is considered epidemiologically unusual. This risk assessment is repeated whenever new information on the event becomes available.

SIGNAL

Signal is data and/or information detected that represent a potential acute risk to human health. Signals may consist of reports of cases or deaths (individual or aggregated), potential exposure of human beings to biological, chemical or radiological and nuclear hazards, or the occurrence of natural or human-made disasters. Signals can be detected through a variety of sources and approaches:

1. **Raw signal:** any information, data or communication collected, obtained, received, or acquired through an initial filtering of

sources used for surveillance activities for the purpose of detecting potential acute risks to human health.

2. **Potential signal:** raw signals that are triangulated with other contemporaneous reports, reviewed against previous trends/occurrences, considered in the context of the situation and dynamically assessed by the PHI analyst in order to retain only those pertinent to early detection purposes. Potential signals are then triaged and either discarded or categorized if they meet predefined criteria.

SOURCE OF INITIAL EVENT REPORTING

It is the source through which the initial information on the event was obtained. Information can be routinely obtained through NFPs or national government channels.

Alternatively, WHO may receive information through detection activities (see: *Detection*) as part of wider public health intelligence operations.

Executive summary

Early warning and response are key to tackling acute public health threats. Detecting health threats rapidly is a crucial element of addressing health emergencies and a core requirement under the International Health Regulations (IHR). It also plays a vital role in the World Health Organization's (WHO) comprehensive emergency response strategy.

Public Health Intelligence (PHI) provides the operational foundation for fulfilling this mandate, enabling the systematic detection, verification, risk assessment and communication of events that may indicate risks to human health.

WHO teams at headquarters, regional and country offices carry out PHI activities daily in close coordination with national authorities and global partners. These activities — aligned with IHR obligations — ensure that health threats are rapidly detected, assessed and reported to enable timely public health actions.

The IHR provide a legally binding framework for 196 States Parties to collaborate in preventing, detecting and responding to public health threats that may affect global populations. At the core of the IHR is the obligation for countries to detect, assess, report and respond to public health events. These obligations also outline the procedures WHO must follow in support of States Parties. To operationalize these responsibilities, PHI plays a pivotal role. Through PHI, WHO and its partners fulfil the IHR mandate by providing timely and actionable intelligence to support evidence-based decision-making in response to emerging or ongoing health emergencies.

This report focuses on the PHI work conducted by WHO in 2024. It presents information and data on WHO's core PHI functions related to event detection, risk assessment and reporting, drawing from multiple internal sources, including: (i) the Event Management System (EMS); (ii) Rapid Risk Assessment (RRA) reports; (iii) the Event Information Site (EIS) for National IHR Focal Points (NFPs); (iv) bulletins; (v) announcements; and (vi) Disease Outbreak News (DON) reports.

In 2024, WHO, through its six regional offices and headquarters, conducted extensive PHI activities, screening approximately 8.5 million raw signals — of which 17 500 were triangulated with other sources of information and assessed to be potential signals — that were then triaged and categorized, leading to the detection of 2805 signals of potential acute risk to human health. This was an increase compared to 2023, reflecting both the strengthening of WHO's PHI systems and the increasing number of acute public health threats globally.

At the same time, the proportion of signals verified within 24 to 48 hours under the IHR continued to increase. Of the 438 verification requests dispatched, 68% were verified under the IHR within 24–48 hours, representing a 6% improvement compared to 2023. This confirms ongoing improvements in early risk assessment and verification workflows over the past five years, highlighting the essential role of States Parties' NFPs. This rigorous surveillance, together with IHR notifications, led to the identification of 432 acute public health events, a figure consistent with the annual averages reported in recent years.

The WHO African Region and the WHO Region of the Americas reported the highest proportions of these events. Over the past five years, these two regions have each consistently contributed between 20–30% of the total events reported globally. Infectious diseases remained the predominant hazard type over the past two decades (2005–2024). In 2024, they accounted for 75% ($n = 322$) of all reported events.

Events due to avian- or animal-origin influenza continued as the most frequent cause of acute public health events in 2024, reflecting the growing global concern surrounding zoonotic spillovers.

The public health relevance of these events is reflected in the information products produced to alert States Parties and the wider public: 39 EIS bulletins focused on avian influenza, accounting for 43% of all bulletins issued. Similarly, DON reports were predominantly issued for events related to zoonotic influenza.

The diversity of health hazards – from zoonotic outbreaks, natural hazards, disasters, armed conflicts to climate-related and product-related events – reaffirms the need for a comprehensive, multi-hazard approach.

Collaboration with platforms such as GLEWS+ – the Global Early Warning System for health threats and emerging risks at the human-animal-environment interface – a tripartite collaboration between the Food and Agriculture Organization, the World Organisation for Animal Health and WHO, together with the integration of regional analyses and risk communication efforts, continue to strengthen WHO's capacity to monitor and respond to emerging threats at the human-animal-ecosystem interface.

In order to save lives and improve public health activities, timeliness is a critical metric. Therefore, evaluating PHI efforts concerning detection, notification, verification and information sharing is key. Between 2020 and 2024, the median time to detect, notify events to WHO and respond to such events, was 7, 2 and 0 days, respectively. These indicators partially meet targets in WHO's 13th General Programme of Work (GPW13) of 7 days for detection, 1 day for notification and 7 days to respond (7–1–7).

During the same period, regional and national RRAs demonstrated median times from initiation to dissemination of 9 and 10 days, respectively. EIS bulletins and DON reports had median publication times of 6 and 5 days, respectively.

Regarding published RRA reports, a decreasing trend was observed, with the number declining from 65 in 2022 to 52 in 2023, and further to 33 in 2024. This decline is attributed to a strategic shift towards conducting more complex regional and global RRAs, rather than focusing on single-country

assessments. The majority of RRAs pertained to events occurring in the African Region, with cholera being the most frequently assessed hazard in 2024. Conversely, the number of EIS bulletins published increased from 73 in 2023 to 90 in 2024. Most EIS bulletins in 2024 addressed events in the WHO Region of the Americas, particularly those related to avian influenza. The number of events for which a DON report was published showed a decreasing trend, from 68 in 2023 to 46 in 2024. The majority of DON reports in 2024 pertained to events occurring in the WHO African Region, with avian influenza being the most commonly reported hazard. In addition, more detailed situation reports were published on mpox, cholera, and the Sudan conflict and refugee crisis.

The 2024 *Global public health intelligence report* provides critical insights into WHO's global PHI activities under the IHR. It highlights improvements in the timeliness of verification requests and indicates that the 7–1–7 target has been achieved for detection and response, although further efforts are needed to enhance the timeliness of notifications to WHO. Another key finding from the report is the prominence of avian influenza in 2024, both as a leading infectious disease hazard and as the primary focus of RRAs, EIS bulletins and DON reports.

In conclusion, this report underscores the vital importance of public health intelligence on a global scale. It emphasizes the pivotal role of the IHR framework in facilitating WHO's engagement with States Parties for the detection, verification, risk assessment and information sharing concerning events that may constitute a public health emergency of international concern. This also highlights the importance of PHI in assessing the risk of an acute public health event or analysis of a situation that eventually activates the internal WHO Emergency Response Framework (ERF).

In a time of global funding constraints, strong public health intelligence systems and coordinated international collaboration remain crucial. Linking local, regional and global efforts is essential to ensure timely detection and response to health threats, thereby protecting global health security.

Background

1.1 Introduction

The International Health Regulations (IHR) provide a legally binding framework for 197¹ States Parties – including 194 World Health Organization (WHO) Member States – to collaborate in preventing, detecting, and responding to public health threats that may affect global populations. At the core of the IHR is the obligation for countries to detect, assess, report and respond to public health events (PHEs). These obligations also outline the procedures that WHO must follow, including detection, assessment and response to potential threats.

To operationalize these responsibilities, Public Health Intelligence (PHI) plays a pivotal role. PHI refers to the systematic process of detecting, verifying, assessing and reporting public health events that could pose a risk to human health. Through PHI, WHO and its partners fulfil the IHR mandate by providing timely, actionable intelligence to support evidence-based decision-making in response to emerging or ongoing health emergencies. PHI operations are carried out daily by dedicated teams at WHO headquarters and across its six regional offices. These teams work closely with WHO country offices, national authorities, National IHR Focal Points (NFPs) and global partners to support the timely identification and management of public health threats in line with the IHR. A full overview of States Parties and their respective WHO regions is provided in Annex 1 of this report.

WHO carries out the four core functions of PHI that are central to the IHR framework (Fig. 1):

- Detection of signals indicating a potential public health threat;

- Verification of these signals to confirm their reliability and relevance;
- Assessment of the associated risks to national, regional and global health security; and
- Alerting/Reporting to disseminate information to enable a timely, coordinated and proportionate response.

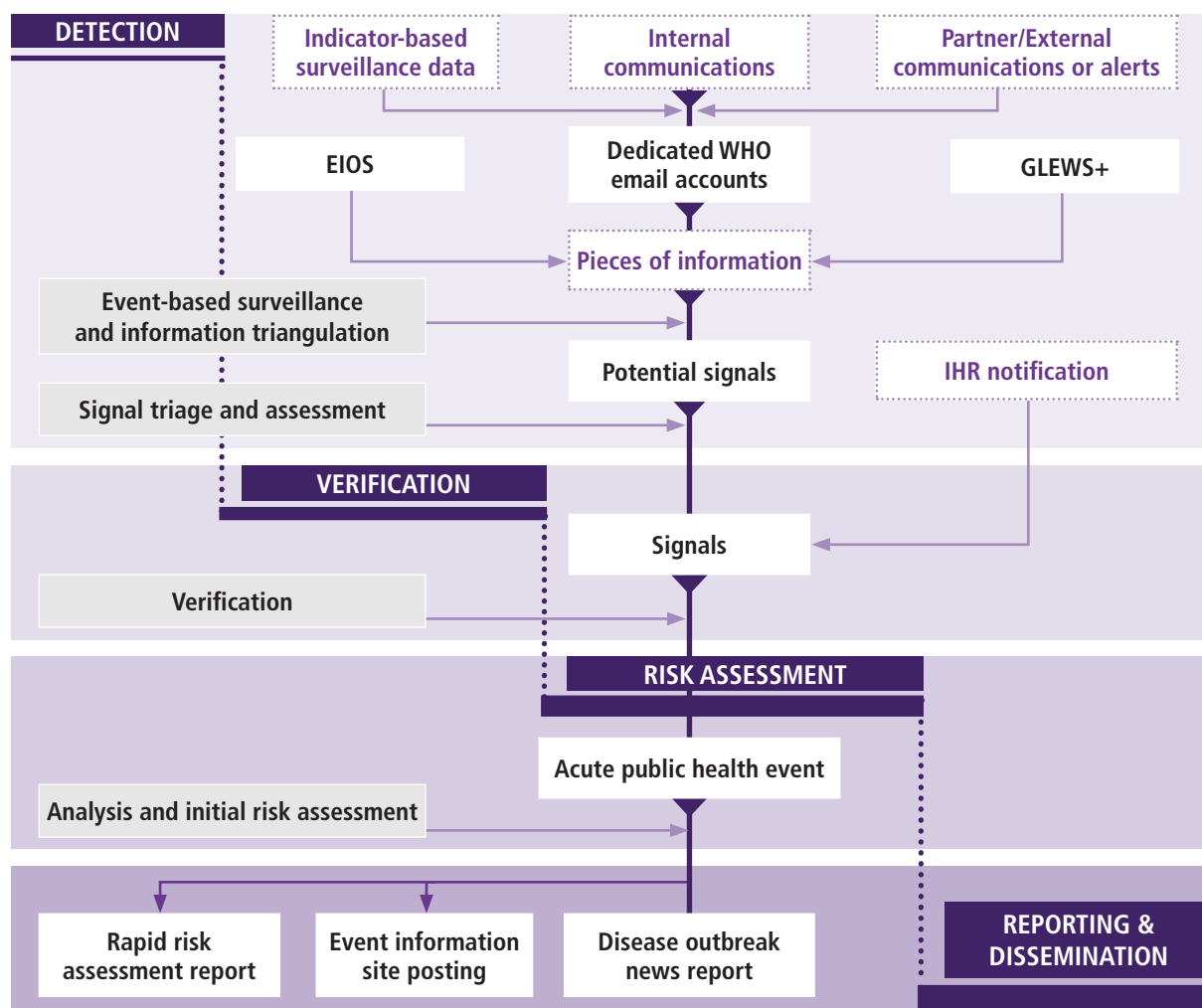
These functions rely heavily on global platforms and collaborative mechanisms to facilitate real-time data collection, analysis and sharing. One of the key platforms is Epidemic Intelligence from Open Sources (EIOS), a multi-stakeholder initiative developed by WHO and partners. EIOS leverages publicly available information, such as news media or official websites, to detect early signals of public health events. It enables rapid scanning of open-source data, helping analysts identify emerging trends and anomalies that might otherwise go unnoticed. Another critical component is the Global Early Warning System for health threats and emerging risks at the human-animal-environment interface (GLEWS+), a tripartite collaboration between the Food and Agriculture Organization of the United Nations, WHO and the World Organisation for Animal Health.

GLEWS+ supports the verification and assessment of zoonotic disease threats by integrating animal and human health surveillance data. It embodies the One Health approach and supports the IHR's requirement to assess risks emerging at the interface of humans, animals and environment.

Together, platforms like EIOS and GLEWS+ contribute to a robust, coordinated intelligence

1. Palestine became a State Party to the IHR on 29 December 2025. As this report refers to 2024, Palestine is not referred to as a State Party to the IHR in the context of this report.

FIG. 1 World Health Organization public health intelligence information flow, associated outputs and activities



Source: Hamblion E, Saad NJ, Greene-Cramer B, et al. (2023) Global public health intelligence: World Health Organization operational practices. PLOS Glob Public Health 3(9):e0002359 (<https://doi.org/10.1371/journal.pgph.0002359>).

network, enabling WHO to meet its IHR obligations. These systems ensure that early warning and risk assessment activities are informed by diverse, multidisciplinary sources of information, including formal surveillance, partner reporting and open-source intelligence.

WHO monitors a wide range of signals that may indicate emerging health threats and include

climate-related signals that are of growing importance, because climate variability and extreme weather events increasingly influence disease patterns, disaster impacts and health system disruptions. Climate-driven hazards are monitored through a number of different platforms, in particular, the Global Disaster Alert and Coordination System (GDACS). The initial assessment of the associated public health risks

constitutes part of the intelligence processes, integrating climate and health data to anticipate cascading effects (for example, floods leading to cholera outbreaks or an increase in vector population). This *Global Public Health Intelligence*

report provides a comprehensive overview of the PHI activities carried out in 2024 by WHO PHI teams worldwide, as part of their mandate to: (i) detect, (ii) verify, (iii) assess and (iv) report public health threats on a global scale.

1.2 Methodology

The methodology of this report is briefly outlined in the following sections: (1) **the data sources** section explains where the data were obtained from and provides additional details about the

sources; and (2) **the data analysis** section summarizes how the data were analysed, including the metrics and tools that were used.

Data sources

Data on signals were obtained from various data management systems used by the WHO regions and headquarters, including the internal WHO signal management application implemented since 2022. Data on the number of signals recorded between 1 January 2024 and 31 December 2024 were provided by the WHO regions, while detailed signal data detected and managed by WHO headquarters for 2024 were extracted from the signal management application.²

Data on events were extracted from the WHO Event Management System (EMS), which is not designed to be a comprehensive database of all new and ongoing acute public health events worldwide, but for events reported to WHO under the IHR. For this report, only data on events classified as substantiated acute public health events were extracted from the EMS. Events were included based on their creation date in the system.

Data on verification requests were obtained from the WHO regional offices. These data were tracked by monitoring requests for verification sent by the WHO regional offices to NFPs and WHO country offices, along with their respective responses between 1 January 2024 and 31 December 2024. The timeliness of responses was evaluated using a benchmark of 48 hours after the request was sent.

Data on the Event Information Site (EIS) bulletins and announcements were extracted from the EIS platform, while data on rapid risk assessment (RRA) reports were retrieved from the EMS. Data on Disease Outbreak News (DON) reports and other information products were obtained from an internal WHO database.

Timeliness metrics related to the detect and respond indicator were provided by the WHO regions, while data on the timeliness of information products were extracted from internal WHO databases.

2. In accordance with resolution WHA78.25 (2025), Indonesia was reassigned to the WHO Western Pacific Region as of 27 May 2025. In this instance, data analysis had already been completed with data pertaining to Indonesia included in the WHO South-East Asia regional aggregates. WHO regional comparisons must be done with caution.

Data analysis

The main findings assessed were centered around signals, verification requests, events, RRA reports, EIS bulletins and announcements, and DON reports. These findings were stratified by year over multi-year periods, with a particular focus on 2024, by WHO region.

For events, results were also assessed based on the channel through which the event was initially reported (either the NFP or national government channels versus WHO or other sources, referred to simply as WHO), as well as by disease, condition or hazard type. Similarly, findings for RRA reports, EIS bulletins, DON reports and other information products were analysed by disease, condition or hazard, and by WHO region. A signal was secondly classified as climate-related if it fulfilled the criteria listed in key WHO guidance documents.^{3,4,5,6,7}

Timeliness measures for events published in the EIS were evaluated using the three components of the *WHO Thirteenth General Programme of Work* (GPW 13): (i) detect (*time to detect*), (ii) notify (*time to notify*) and (iii) respond (*time to respond*) indicators. These components are based on approximate event milestones (event start date, event detection date, date event was notified to WHO and date response measures

were implemented) to estimate the median and interquartile (IQR) ranges in days. Negative values (for example, if the response began before the event was notified to WHO) were set to zero.

- **Time to detect** is the period between the event start and detection dates;
- **Time to notify** is the period between event detection and notification to WHO; and
- **Time to respond** is the period between event detection and the implementation of response measures.

Timeliness for information products (RRAs, EISs and DONs) was calculated as the time between when the first draft was started and the date of publication.

Data were presented as averages, medians or aggregated values. Tables and figures were created using RStudio (version 1.3.1093, Boston, MA, United States of America).

A double data analysis was also conducted to ensure consistency and quality. Previous and future reports may show minor discrepancies due to routine updates and cleaning of signal, verification request, EMS, RRA, EIS and DON data.

3. Climate change and health. Geneva: World Health Organization; 2023 [cited 11 Dec 2025] (<https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health>).
4. Operational framework for building climate resilient health systems. Geneva: World Health Organization; 2015 [cited 11 Dec 2025] (<https://www.who.int/publications/i/item/9789241565073>).
5. Protecting health from climate change: vulnerability and adaptation assessment. Geneva: World Health Organization; 2013 [cited 11 Dec 2025] (<https://www.who.int/publications/i/item/9789241564687>).
6. Climate and health country profiles 2015: a global overview. Geneva: World Health Organization; 2015 [cited 11 Dec 2025] (<https://www.who.int/publications/i/item/WHO-FWC-PHE-EPE-15.01>).
7. WHO guidance for climate-resilient and environmentally sustainable health-care facilities. Geneva: World Health Organization; 2020 [cited 11 Dec 2025] (<https://www.who.int/publications/i/item/9789240012226>).

Main findings

2.1 Detection and verification of acute public health events

Detection

In 2024, PHI activities conducted across WHO headquarters and the six WHO regional offices screened close to 8.5 million raw signals, of which 17 500 (0.2%) were triangulated with other sources of information and assessed to be potential signals that were then triaged and categorized, leading to the detection of 2805 (0.2%) signals of potential acute risk to human health (Fig. 2). This represents a 91% increase compared to 2023, when 1470 signals were detected by dedicated teams at WHO headquarters and all regional offices.

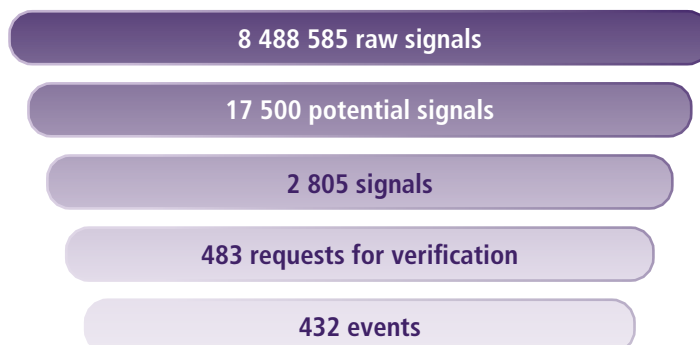
At WHO headquarters, among the 590 signals detected in 2024 (an increase of 10% compared to 2023 when 533 signals were detected), infectious hazards make up the vast majority with (483; 82%), followed by product-related hazards (40; 7%), disaster-related hazards (33; 6%) and chemical hazards (9; 1%). Other hazards (including

animal, food safety, non-communicable, undetermined and zoonosis) collectively account for 4% ($n = 25$) (Fig. 3).

Of the 590 all-hazard-related signals detected at WHO headquarters, 51% could be classified as climate-related signals. The WHO African Region accounted for the largest proportion, with (26%) of the total, closely followed by the WHO Region of the Americas (25%), the WHO Western Pacific Region (18%), the WHO South-East Asia Region (12%), the WHO European Region 9% and the WHO Eastern Mediterranean Region) 8%. Multi-regional signals overall represent 2% of the total.

Climate change related signals include: natural hazards such as geophysical events like earthquakes and volcanic eruptions; meteorological hazards like severe storms, extreme winds, and

FIG. 2 Number of raw signals, potential signals, signals, request for verification and events, 2024

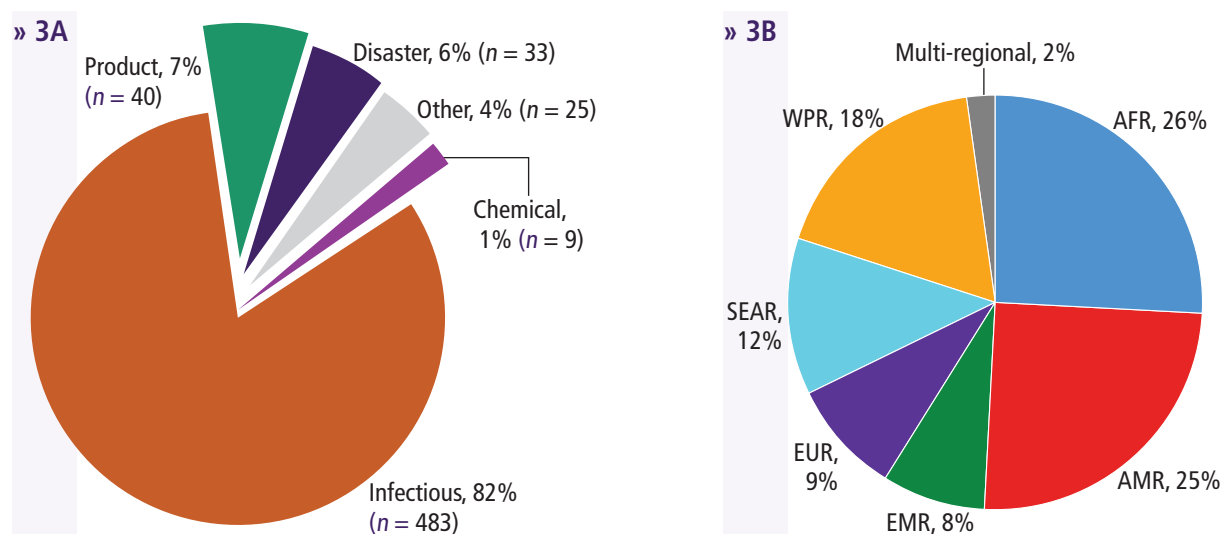


Note: not all events are detected as signals initially, some are reported directly from Member States as IHR notifications, as shown in figure 1.

extreme temperature; hydrological threats such as floods; climatological hazards such as droughts and wildfires, which are increasingly influenced by

long-term climate change; and biological hazards, such as some infectious disease outbreaks including arboviruses.

FIG. 3 Proportion of signals detected at WHO headquarters, by hazard category (A), and all hazard climate-related signals detected at WHO headquarters by region of occurrence (B), 2024 (in %)



WHO regions: AFR, WHO African Region; AMR, WHO Region of the Americas; EMR, WHO Eastern Mediterranean Region
Source: WHO. EUR, WHO European Region; SEAR, WHO South-East Asia Region; WPR, WHO Western Pacific Region

Verification of signals

The process of signal detection – from initial selection of potential signals through triangulation, triage and categorization of signals – is conducted by PHI analysts, a group of epidemiologists with multidisciplinary backgrounds. After triage and categorization, verification requests are sent for those signals for which verification by Member States was required (signals for verification). In 2024, 438 requests for verification were sent by WHO regional and country offices.

Overall, 68% of signals for verification were verified under the IHR within 24–48 hours, which represents an increase by 6% compared to 2023, when 62% of requests for verification were received within 48 hours of the verification request (Fig. 4). Over the past five years, a trend towards improving

the timeliness of response for verification requests has been observed.

The number of requests for verification sent differed between WHO regions. This is in part due to regional differences in the underlying health threats that occur, and partly due to procedural and information management practice. In 2024, the WHO Eastern Mediterranean Region reported the highest number of requests for verification sent (179), followed by the WHO African Region (150), the WHO Region of the Americas (43), the WHO South-East Asia Region (38), the WHO Western Pacific Region (17) and the WHO European Region (11). The proportion of response received within 48 hours ranged from 37% in the the WHO South-East Asia Region to 94% in the WHO Western Pacific Region (Fig. 5).

FIG. 4 Number of requests for verification (A) and response to verification requests (B) sent by response time, 2020–2024 (in %)

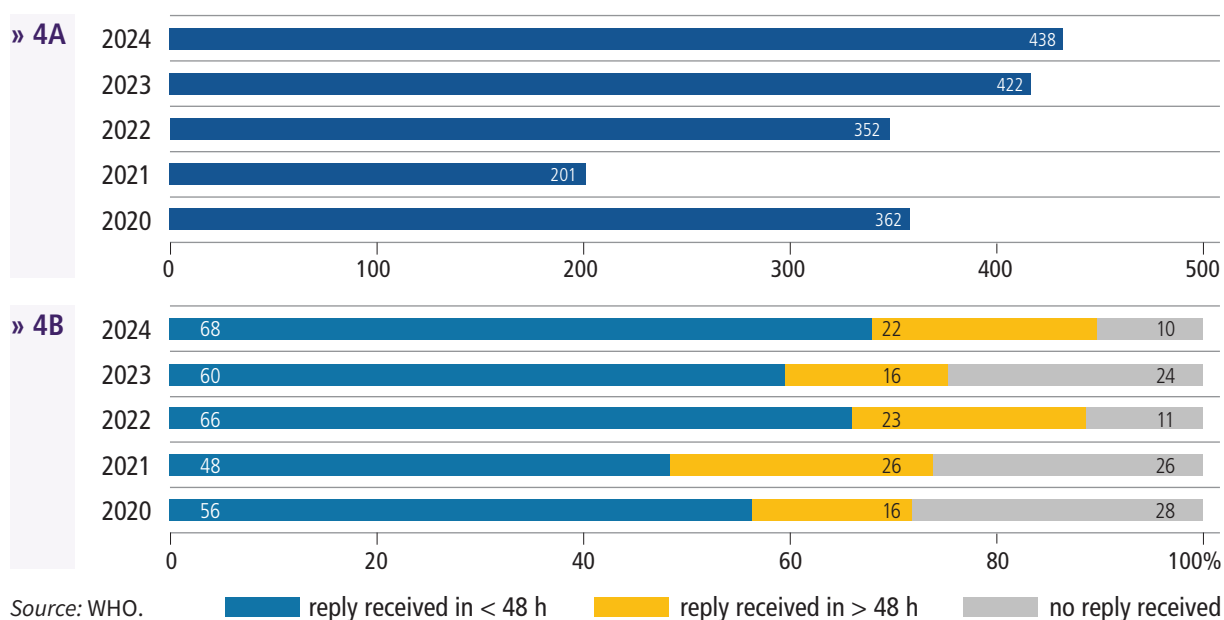
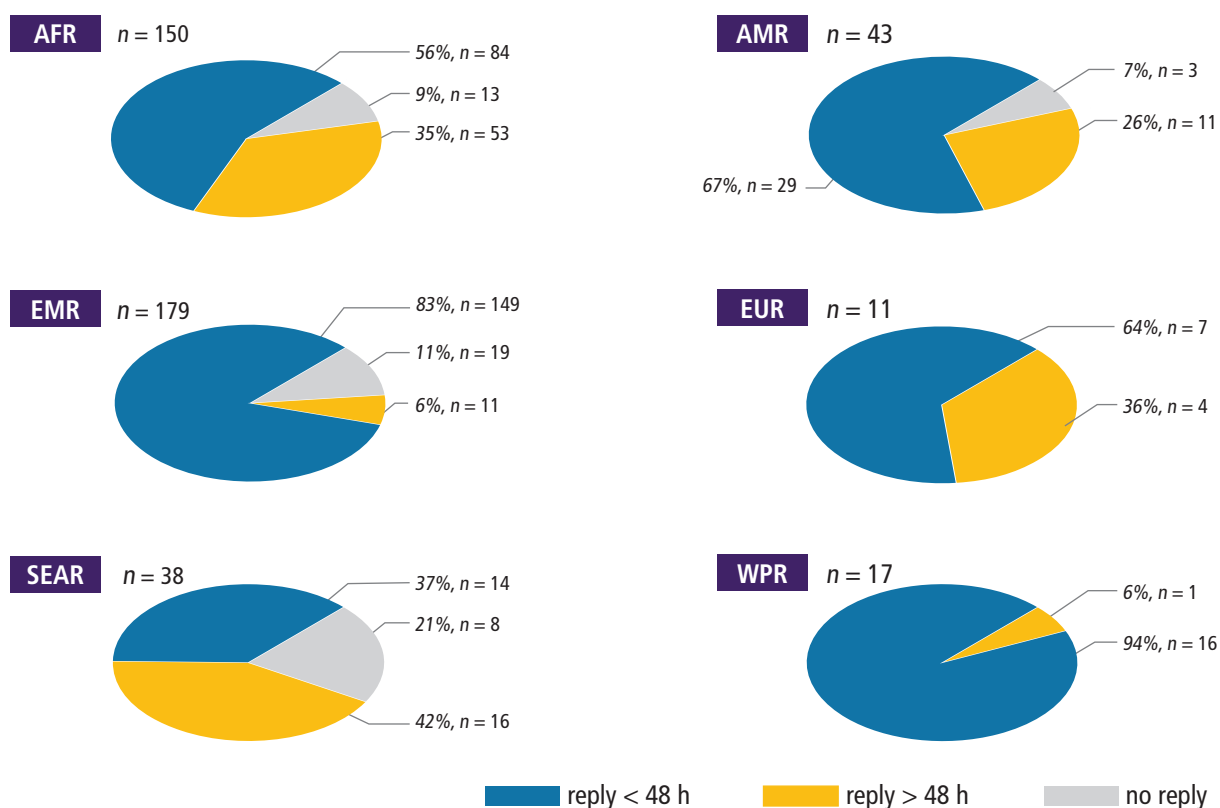


FIG. 5 Response time to verification requests sent, by WHO region, 2024 (in %)



WHO regions: AFR, WHO African Region; AMR, WHO Region of the Americas; EMR, WHO Eastern Mediterranean Region
Source: WHO. EUR, WHO European Region; SEAR, WHO South-East Asia Region; WPR, WHO Western Pacific Region

2.2 Acute public health events

In this section, all findings refer to events that have been verified and designated as substantiated, which are events where the presence of a hazard is confirmed or the number of human cases exceeds established thresholds.

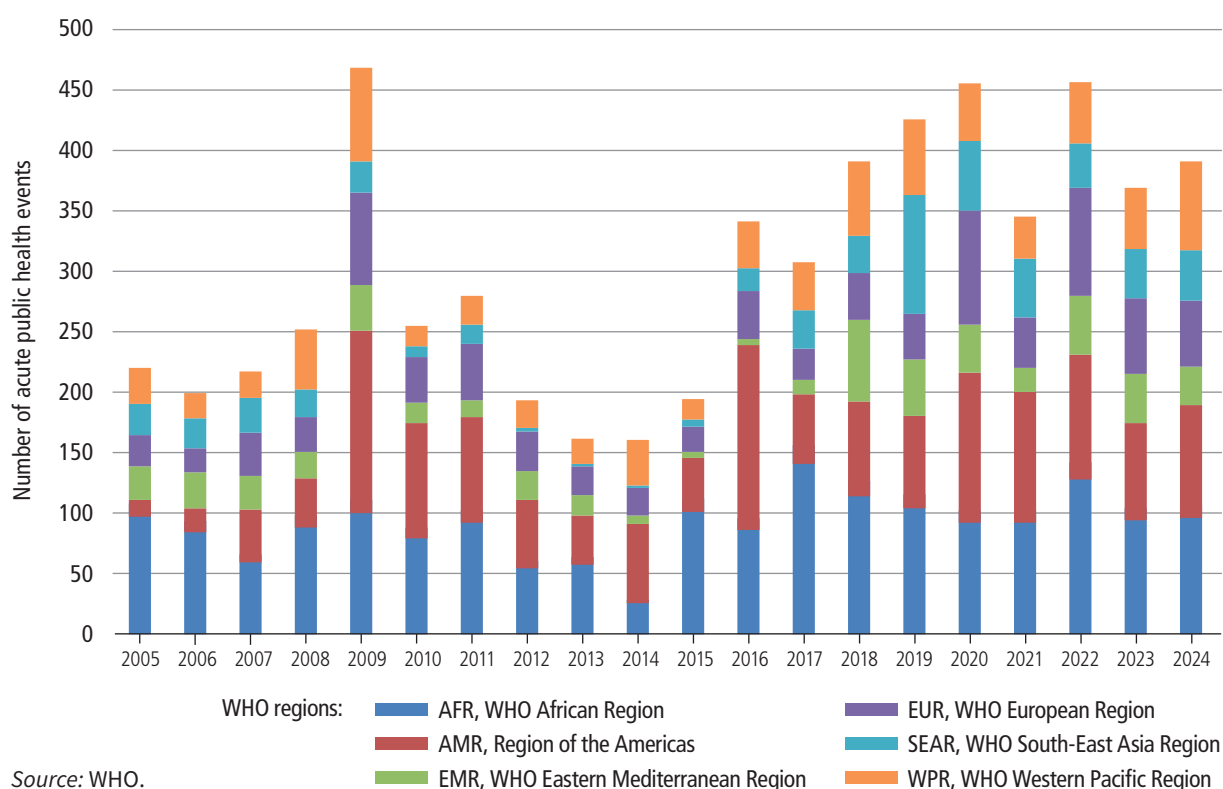
To facilitate readability, they are referred to as “acute public health events”, or simply “events”, rather than “verified and substantiated acute public health events”.

Acute public health events by year and WHO region

In 2024, there were 432 acute public health events reported globally. While this is an increase compared to the previous year, it is in line with the number of events reported in recent years (Fig. 6).

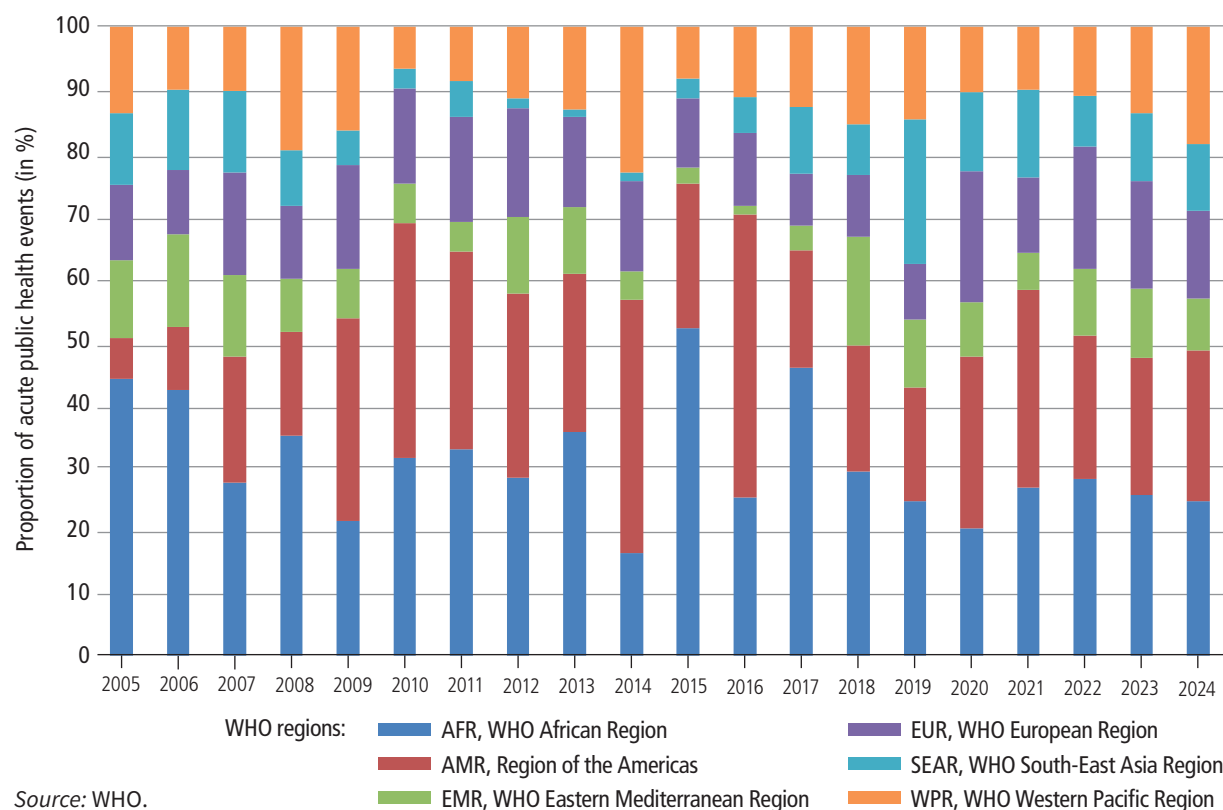
Over the past 20 years (2005–2024), 6052 events were reported; the number of events per year ranged from 159 to 463, with a median of 301 events.

FIG. 6 Acute public health events by WHO region, 2005–2024 (in absolute numbers)



The reported number of acute public health events differed across WHO regions. In 2024, the WHO African Region and the WHO Region of the Americas reported the highest proportions of acute public health events: $n = 141$ [32.6%] and $n = 92$ [21.3%], respectively (Fig. 7).

Over the past five years, the events reported from each of these two regions have represented roughly 20–30% of the total events reported, whereas the events occurring in each of the other regions have typically varied between 5% and 20%.

FIG. 7 Proportion of acute public health events by WHO region, 2005–2024, (N = 6060) (in %)

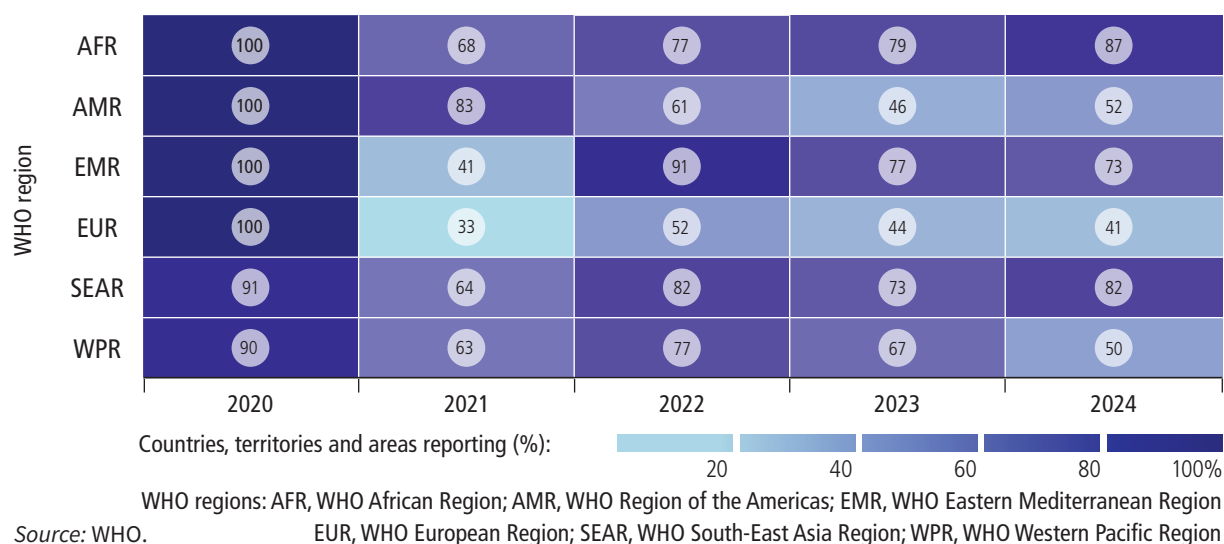
Different elements contribute to the variations in the number of events reported by each region, including the number of countries, territories and areas within those regions; geographical location, economic conditions, political instability, climate change, population density and health infrastructure, to name a few.

The analysis of the proportion of countries, territories and areas that have reported acute public health events shows that from 2020 to 2024, all the regions reached almost 100% of country, territory and area reporting at least one acute public health event, corresponding to the global pandemic period (Fig. 8). Following this period, there is a noticeable divergence in reporting across regions. The WHO European and Western Pacific regions showed a substantial decline in the proportion of areas reporting events, whereas others maintained relatively higher reporting percentages.

In 2024, at least two-thirds of countries, territories and areas within three regions reported an event, without any discernible pattern associated with the size of the region or the frequency of events. It is to note that these data pertain solely to newly reported acute public health events and do not encompass information regarding ongoing public health events that are still under monitoring.

Between 2023 and 2024, a decrease in the proportion of countries, territories and areas where an event was reported in the WHO Western Pacific Region, dropped from 67% to 50%; in contrast, the WHO South-East Asia Region showed an increase from 73% to 82%. These shifts may reflect changes in surveillance sensitivity, reporting practices or true variation in the occurrence of acute public health events across the regions.

FIG. 8 Proportion of countries, territories and areas reporting at least one acute public health event, by WHO region, 2020–2024 (in %)

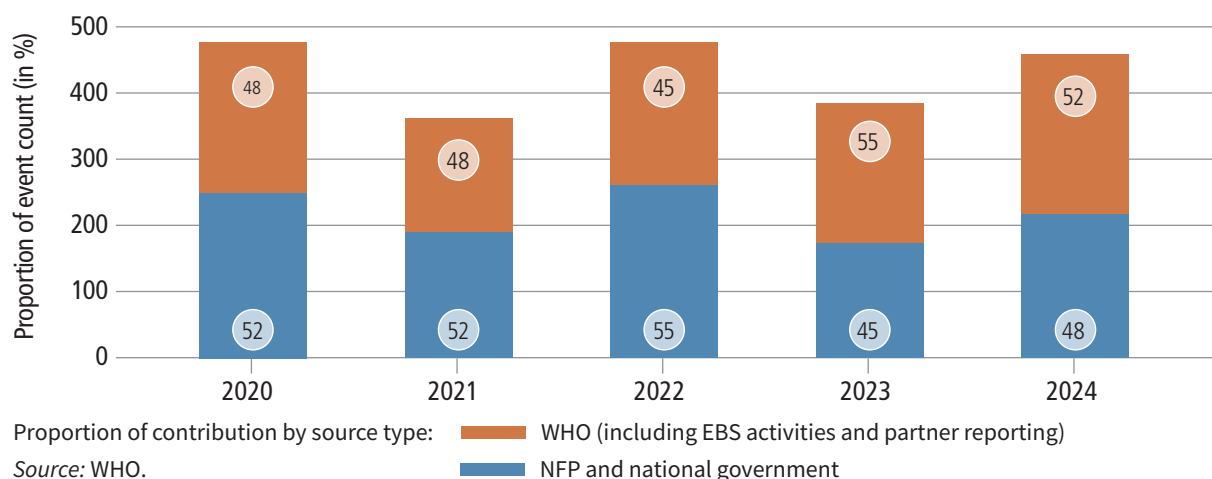


Initial source of event reporting

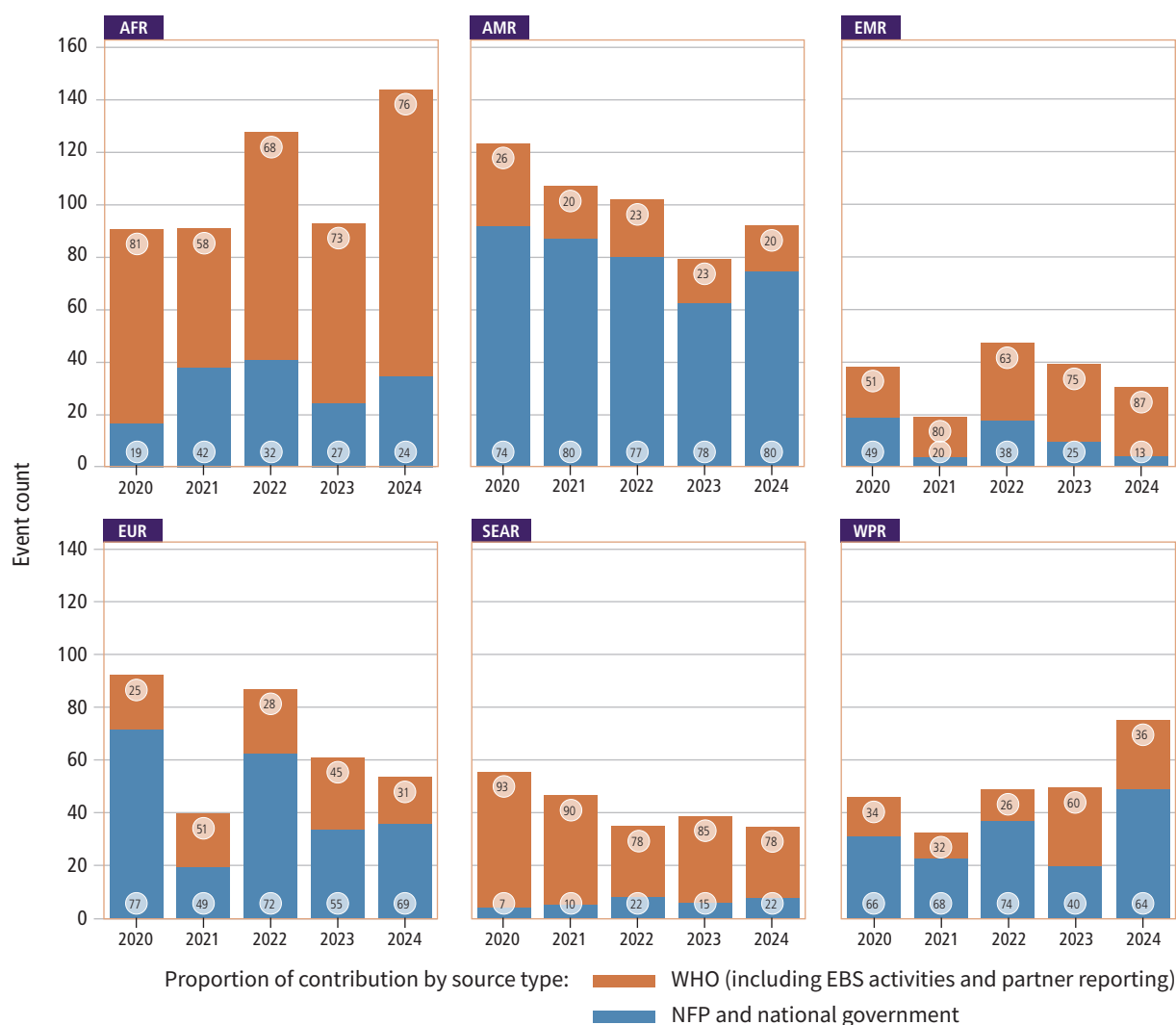
Between 2020 and 2024, for events monitored by WHO, the initial source of information has equally been by IHR NFPs (or through official national government channels) and through other sources identified by WHO (Fig. 9). Differences in the initial source of event reporting are observable at the regional level (Fig.10).

In the WHO African, Eastern Mediterranean and South-East Asia regions, there is a discernible trend where events are primarily identified by WHO rather than reported by IHR NFPs. Conversely, in the WHO Region of the Americas, the WHO European and Western Pacific regions, the majority of events have been initially reported by IHR NFPs during the five-year period 2020–2024.

FIG. 9 Initial source of information for events, 2020–2024



The data for initial source of information for 2023 in the GPHIR 2024 have been retrospectively adjusted from 157 (43%) to 165(45%) for the NFP and national government.

FIG. 10 Initial source of information for events by WHO region, 2020–2024

WHO regions: AFR, WHO African Region; AMR, WHO Region of the Americas; EMR, WHO Eastern Mediterranean Region
 Source: WHO. EUR, WHO European Region; SEAR, WHO South-East Asia Region; WPR, WHO Western Pacific Region

Hazard type of acute public health events

During the past 20 years (2005–2024), infectious diseases have remained the predominant hazard type for events globally (Fig. 11). In 2024, infectious diseases accounted for 75% (322) of all events globally, while the second most common cause were disasters (16%, 68). Of note, beginning in 2022, food safety and animal/zoonosis hazards were reclassified to other respective hazards, and therefore, explain the absence of these hazard types in 2023 and 2024.

Across regions, while infectious diseases remain dominant, disaster-related events contributed on average about 22% of the yearly events in the WHO South-East Asia Region and 9% in the WHO Western Pacific Region (Fig. 12). Similarly, food safety issues accounted for roughly 11% of annual events in the WHO European Region and 5.5% in the WHO Region of the Americas. Additionally, animal/zoonosis hazards made up about 7.5% of the yearly events in the WHO Western

Pacific Region and 7.3% in the WHO South-East Asia Region, whereas product-related incidents contributed around 5.7% of annual events in the WHO Region of the Americas.

To examine the most common diseases or conditions of an infectious nature, the ten most frequently reported diseases or conditions of an infectious nature were identified across the five-year period 2020–2024 (Fig. 13).

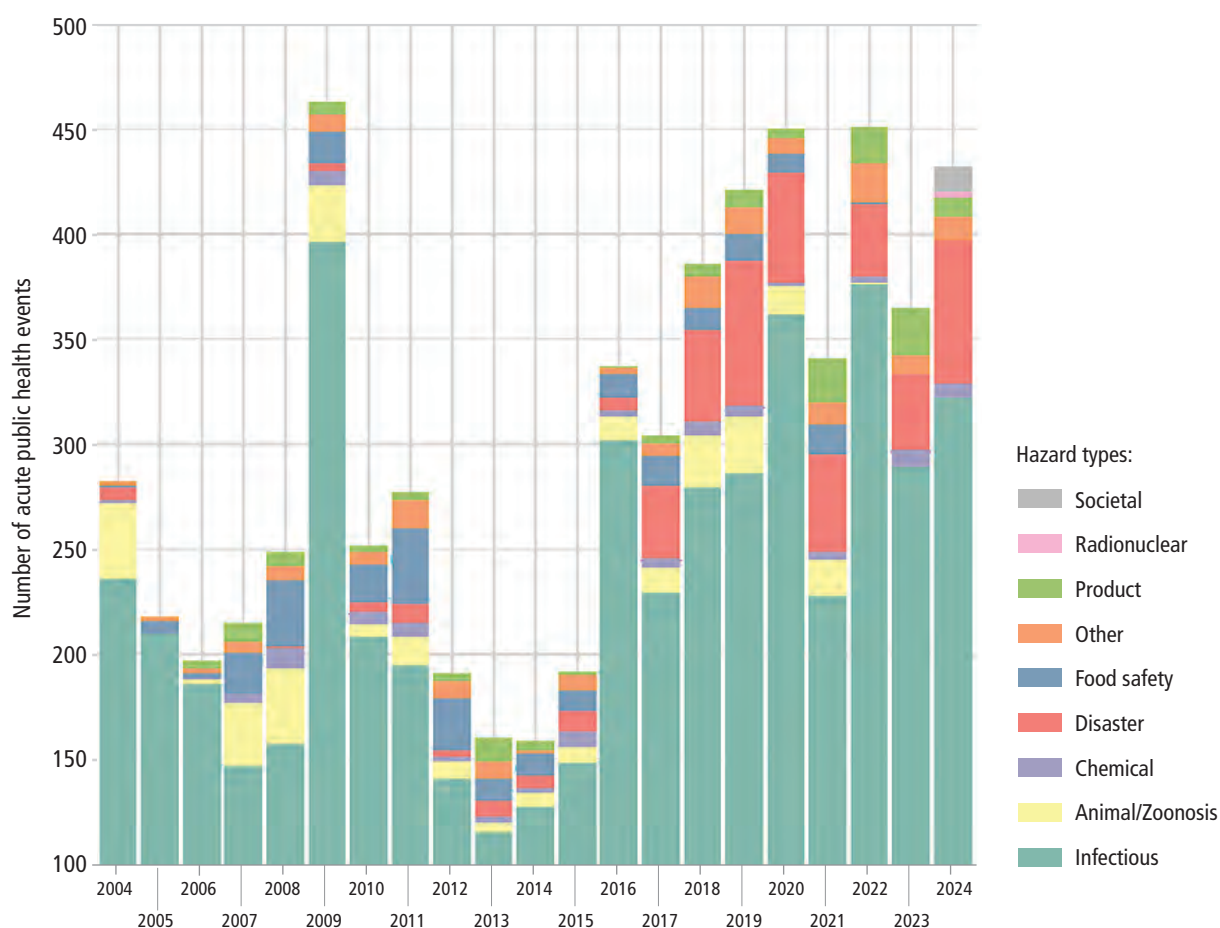
In the past five years, ten distinct infectious diseases or conditions have been the most frequent causes of acute public health events each year. Notably, most of the ten most frequently reported diseases or conditions of an infectious nature appeared consistently every year. Aggregated across the five-year period, new infectious

disease events were dominated by COVID-19 ($n = 278$), followed by mpox ($n = 121$), dengue ($n = 107$), cholera ($n = 100$), avian/animal influenza ($n = 92$) and measles ($n = 85$).

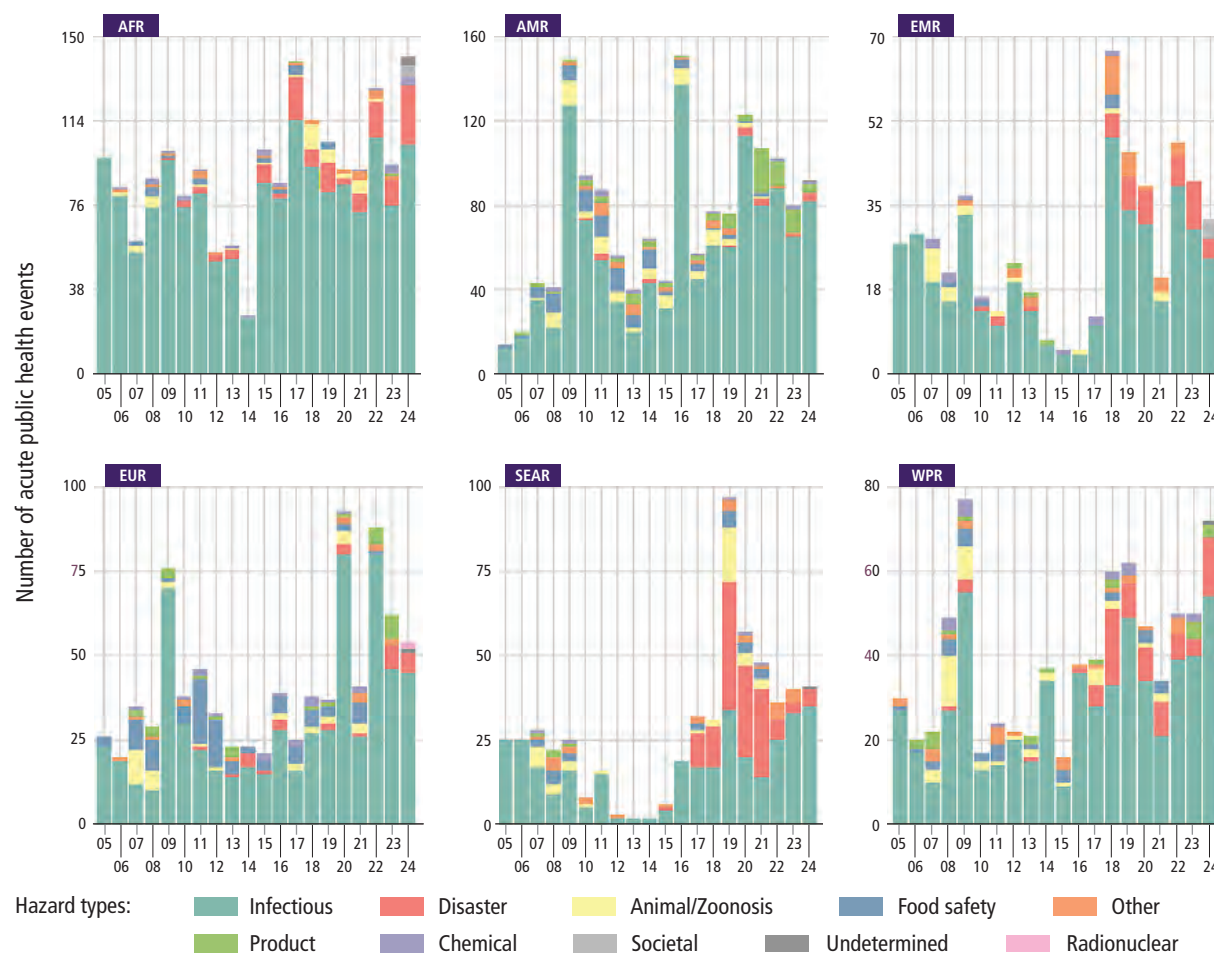
In 2024, influenza due to avian or animal origin emerged as the leading cause of infectious disease events ($n = 35$), followed by dengue and measles (each with 23 events), poliomyelitis ($n = 21$) and mpox ($n = 19$).

This represents a shift from the earlier years of the period, where COVID-19 had dominated the landscape of new infectious disease events. Figure A (Annex 2) provides a detailed analysis of the five most frequently reported diseases, conditions or hazards of an infectious nature, by WHO region across the five-year period 2020–2024.

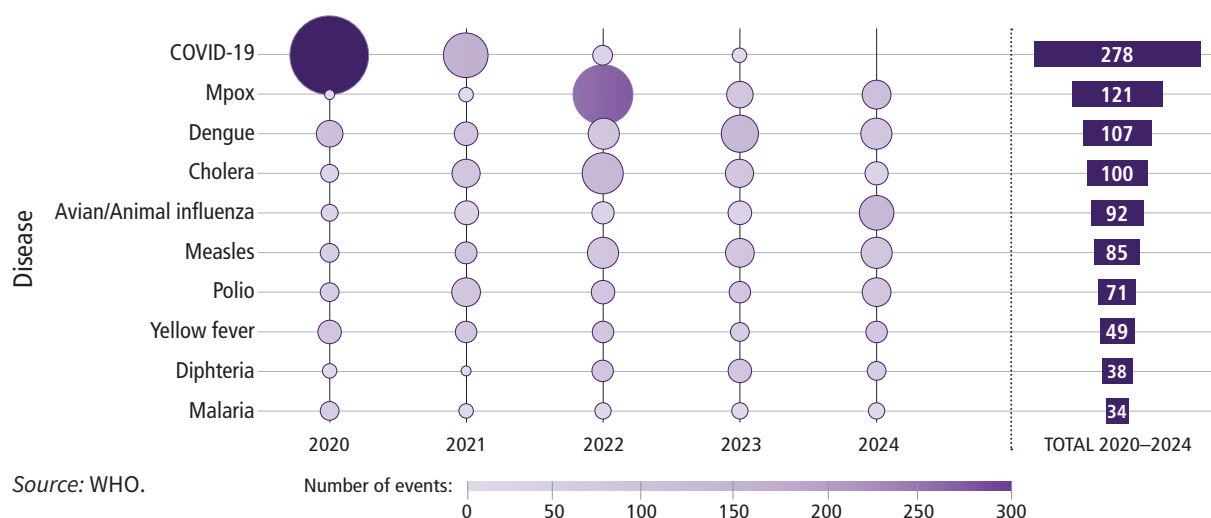
FIG. 11 Acute public health events by hazard type, 2005–2024



Source: WHO.

FIG. 12 Acute public health events by hazard type and WHO region, 2005–2024 (05–24)

Source: WHO.

FIG. 13 Acute public health events for the 10 most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024

2.3 Rapid risk assessment of acute public health events

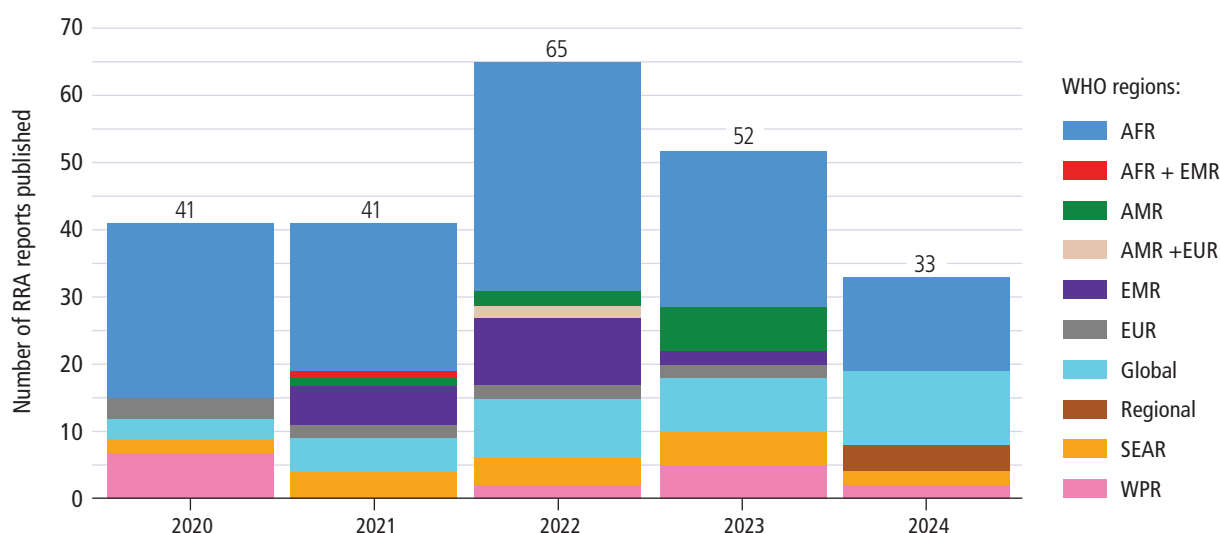
Between 2020 and 2024, 232 RRA reports were published. Following an increase of RRAs published in 2022 (65), there has been a decrease in the number of total RRAs annually to 33 in 2024 (Fig. 14). This reflects a shift in rapid risk assessment priorities toward more resource-intensive regional and global assessments, with high technical complexity.

Almost half of the total number of RRA reports published over the past five years were related to acute public health events reported solely in the WHO African Region (47.81%, 111/232). This was followed by reports addressing global

acute public health events (15.9%, 37/232), and those for events occurring in the WHO Eastern Mediterranean Region (11.6%, 27/232).

The number of RRAs on events occurring in a single country represented approximately 75% (175/232) of all RRAs conducted annually in the past 5 years, with 93% (38/41) in 2020, 78% in 2021 (32/41) and 2022 (51/65) respectively, 71% (37/52) in 2023 to 52% (17/33) in 2024. Notably, an increase in the number of global RRAs conducted annually has been observed over the past five years, with their proportion growing from 7% of total RRAs conducted in 2020 to 33% in 2024.

FIG. 14 Number RRA reports published by year and geographical area, 2020–2024 (in absolute number)



WHO regions: AFR, WHO African Region; AMR, WHO Region of the Americas; EMR, WHO Eastern Mediterranean Region
Source: WHO. EUR, WHO European Region; SEAR, WHO South-East Asia Region; WPR, WHO Western Pacific Region

Rapid risk assessment reports by hazard, disease or condition

During the 2020–2024 period, 98% (228/232) of the RRAs performed were related to an infectious hazard. The remaining 2% (4/232) involved acute public health events for environmental contamination ($n = 1$), the risk of a biological hazard from

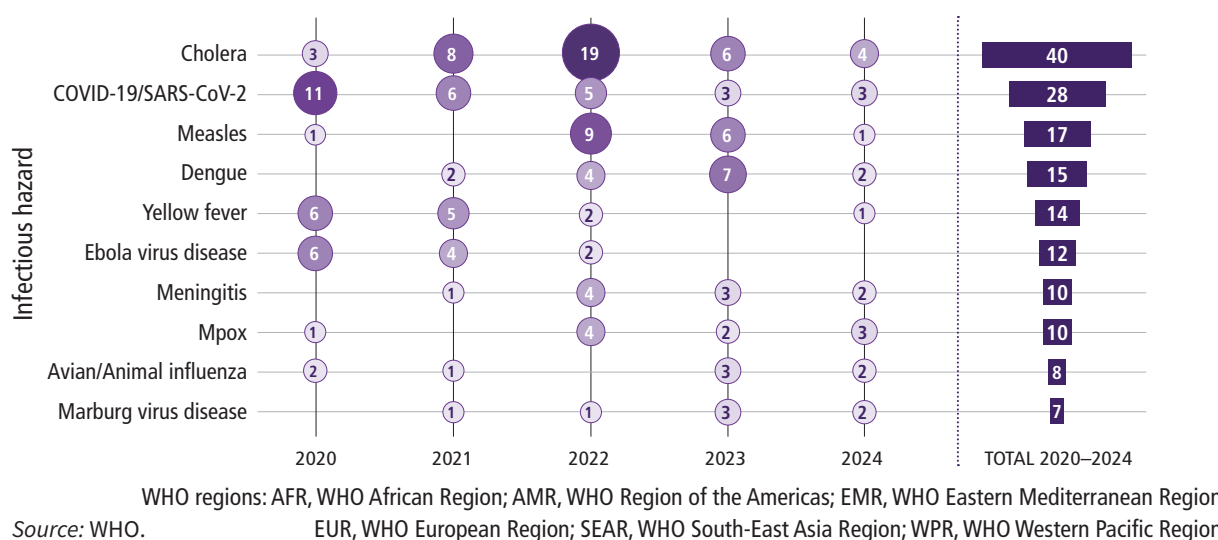
a seized laboratory ($n = 2$) and a chemical event ($n = 1$).

Figure 15 shows the 10 most common infectious hazards for which RRA reports were published

between 2020 and 2024 (Fig. 15). Cholera emerges as the dominant infectious hazard assessed over the five-year period, accounting for 16.8% of the total with 40 RRAs, followed by COVID-19 with 11.8% (28 RRAs) and characterized by a peak in 2020 during the initial phase of the pandemic. The temporal analysis reveals that while COVID-19 assessments were concentrated in the early years, particularly 2020, cholera assessments

have shown an increasing trend, suggesting evolving public health priorities and emerging challenges. Figure B in Annex 2 provides the number of RRAs by WHO region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024.

FIG. 15 The 10 most frequent public health events of an infectious nature for which a RRA was conducted, for each year of the past five years and aggregated across the five-year period, 2020–2024



Outcome of rapid risk assessment by risk level

RRAs assess the risk of a public health event at various geographical scales. For example, a single country RRA will assess the risk at the national, regional and global levels, while a regional RRA will assess the risk at regional and global levels, and global RRA will assess the risk at a global scale.

Among single-country RRAs (175/232, 75%), the majority of events were classified as high risk at the national level (57.7%), followed by very high risk (20.6%). The proportion of assessments with moderate or low national risk levels was substantially lower (16.6% and 5.1%, respectively).

This suggests a prioritization of events for conducting a RRA with a potentially significant impact nationally.

In terms of regional risk, single-country events were most frequently classified as having a moderate regional risk (50.9%), with fewer assessments rated as high (17.1%) or low (26.9%). Only 5.1% were considered very high at the regional level. The public health risk at the global level of a single country event was classified as low for nearly all events (94.9%) (Table 1).

For regional and multi-regional events, half (50.0%) of RRAs were assessed as high risk at the regional level and 40.0% as moderate. At the global level, over half of these events were considered low risk (65.0%). Among the events for which global RRAs was conducted, 40.5% were rated

very high risk at the global level, 27.0% as high, and 21.6% as moderate, indicating that events for which a globally scoped RRA is conducted are more likely to be associated with significant international public health implications.

TABLE 1 Risk level by type (global, multi-regional, regional or single country) for RRA reports published, 2020–2024

Country/Regional/Global	National risk level	Regional risk level	Global risk level
Single country (N = 175)			
» Very high	36 (21%)	9 (5%)	4 (2%)
» High	101 (58%)	30 (17%)	3 (2%)
» Moderate	29 (17%)	89 (51%)	2 (1%)
» Low	9 (5%)	47 (27%)	166 (95%)
Regional and multi-regional (N = 20)*			
» Very high	—	—	—
» High	—	10 (50%)	—
» Moderate	—	8 (40%)	2 (10%)
» Low	—	2 (10%)	13 (65%)
Global (N = 37)**			
» Very high	—	—	15 (41%)
» High	—	—	10 (27%)
» Moderate	—	—	8 (22%)
» Low	—	—	—

* Five regional RRA reports only assigned a regional, but not an overall global risk level.

** Four global RRA reports have an overall level of risk established by risk groups and/or clade, and were not included in the count.

2.4 Global information dissemination for acute public health events

EIS bulletins and announcements

Between 2020 and 2024, there were 524 EIS bulletins published, with an average number per year of 105 bulletins. The number of EIS bulletins was the highest in 2020, due to the COVID-19 pandemic, then declined, reaching the fewest in 2023. In 2024, the number of EIS bulletins increased by 23% compared to the previous year with

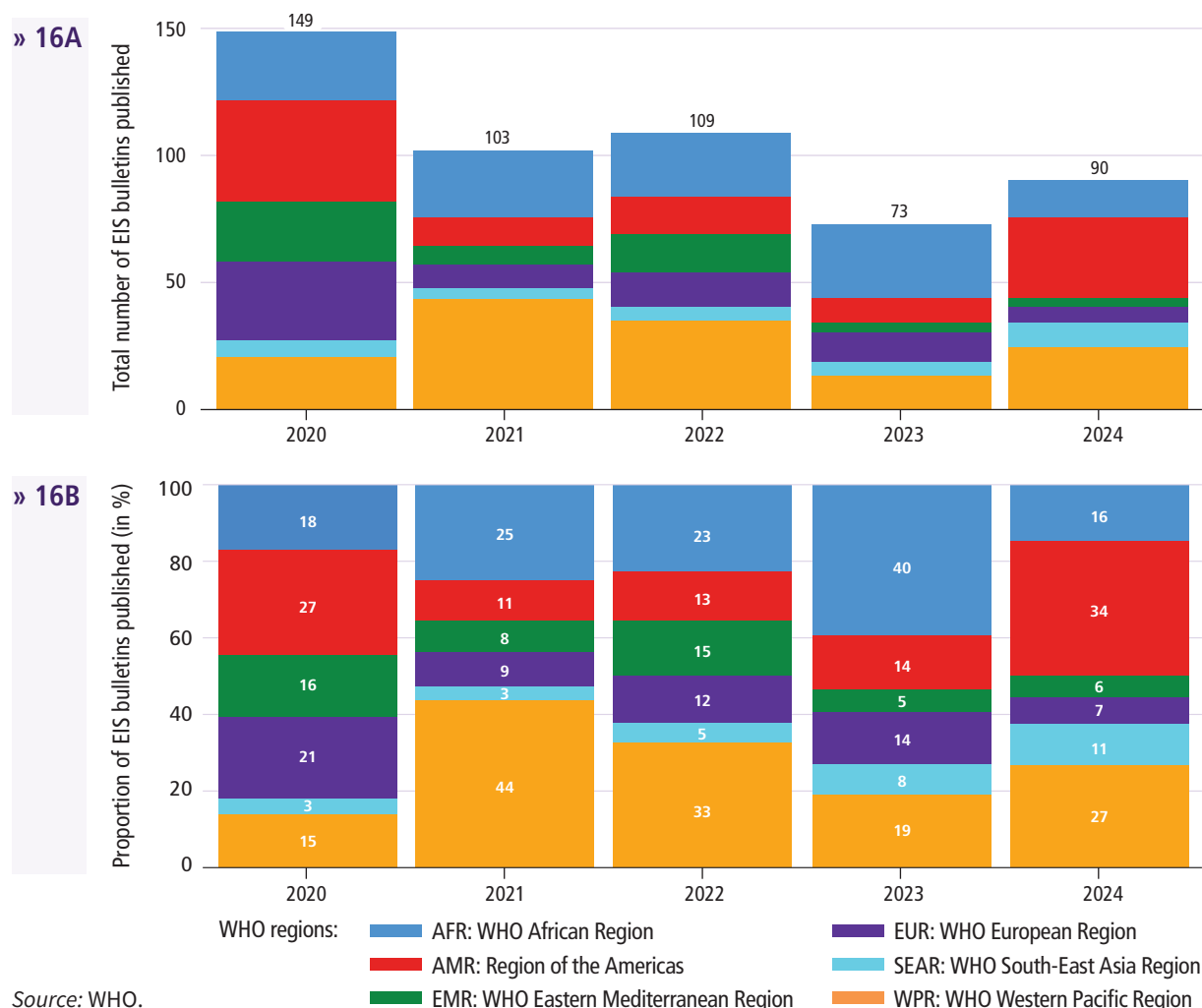
90 bulletins published, mostly covering avian/animal influenza events in the WHO Region of the Americas and in the WHO Western Pacific Region (Fig. 16). In addition, in 2024, a total of 25 EIS announcements were also published that represents a decrease compared to 2023, when 52 EIS announcements were published.

More than a quarter of EIS bulletins published during the past five years (2020–2024) were about public health events in the WHO Western Pacific Region, which accounts for 27% (141/524) of all bulletins. This is closely followed by the WHO African Region with 23% (121/524) of bulletins published. Bulletins on events in the WHO Region of the Americas represent 20% (106/524) of the total, while those on events in the WHO European Region comprise 13% (69/524). The WHO Eastern Mediterranean and South-East Asia regions have lower proportions at 11% (57/524) and 6% (30/524), respectively.

The annual distribution of EIS bulletins varied across WHO regions. In 2020, the distribution

was relatively balanced among the regions, with the WHO Region of the Americas showing the highest proportion at 27% (40 out of 149 bulletins). However, in 2021 and 2022, the WHO Western Pacific Region accounted for the largest proportion (44% and 33% of the total bulletins, respectively), driven by events of avian/animal influenza. In 2023, the WHO African Region showed the highest proportion (40%, 29 out of 73 bulletins), and in 2024, the WHO Region of the Americas once had the largest percentage with 34.4% (31 out of 90 bulletins), primarily driven by the avian influenza events in the region. In 2024, the WHO South-East Asia Region also recorded the highest number of EIS bulletins published in this region in the past five years.

FIG. 16 EIS bulletins published by year and region, in absolute number (A) and in percentage (B), 2020–2024



Source: WHO.

Excluding one EIS bulletin, which was published on floods in the WHO Eastern Mediterranean Region, all EIS bulletins during the five-year period pertained to infectious hazards.

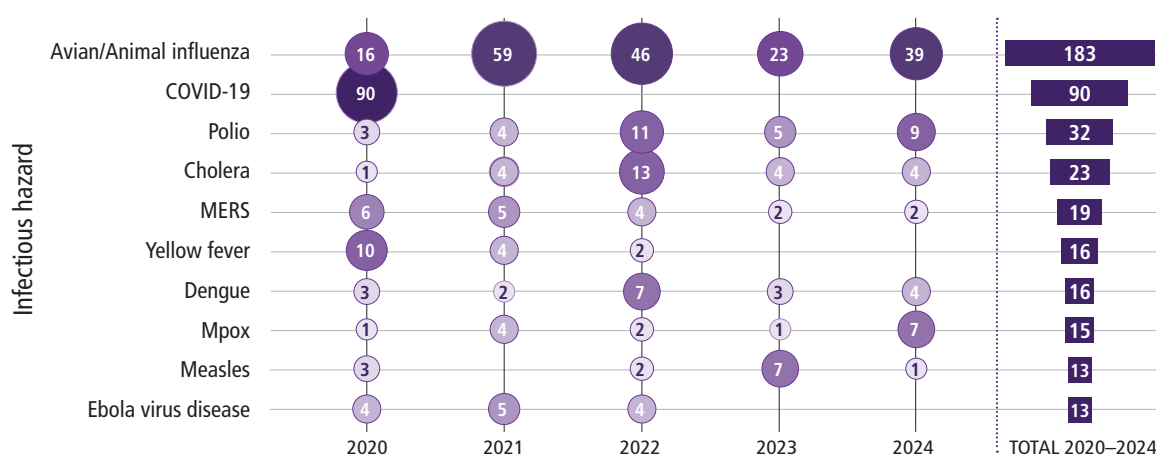
Figure 17 shows the ten most common infectious diseases for which an EIS bulletin was published (Fig. 17).

Aggregated across the five-year period, the EIS bulletins on infectious disease events were dominated by influenza due to identified avian or animal influenza virus (183 events, 35% of total), followed by COVID-19 (90 events, 17%), polio (32 events, 6%) and cholera (23 events, 4%). The temporal distribution reveals significant shifts in the infectious disease landscape. In 2020, COVID-19 represented the majority of bulletins published (90 bulletins), accounting for approximately 60% of that year's reports.

By 2021, while there were no longer bulletins published on COVID-19 (as alternative methods of reporting were employed such as dedicated situation reports and dashboards), other infectious hazards like influenza due to identified avian or animal influenza virus (59 events, 57.3%) gained prominence, along with Ebola virus disease and Middle East respiratory syndrome coronavirus (5 events each, 5%). In 2024, bulletins on events of influenza due to identified avian or animal influenza virus were mostly published, with 39 bulletins, representing 43% of all bulletins published this year. Other events with significant number of bulletins in 2024 include polio (9 bulletins, 10%) and mpox (7 bulletins, 8%).

Figure C in Annex 2 provides a detailed analysis of the five most frequently reported diseases, conditions or hazards of an infectious nature published in EIS bulletins by region across the five-year period 2020–2024.

FIG. 17 EIS bulletins published for the ten most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024



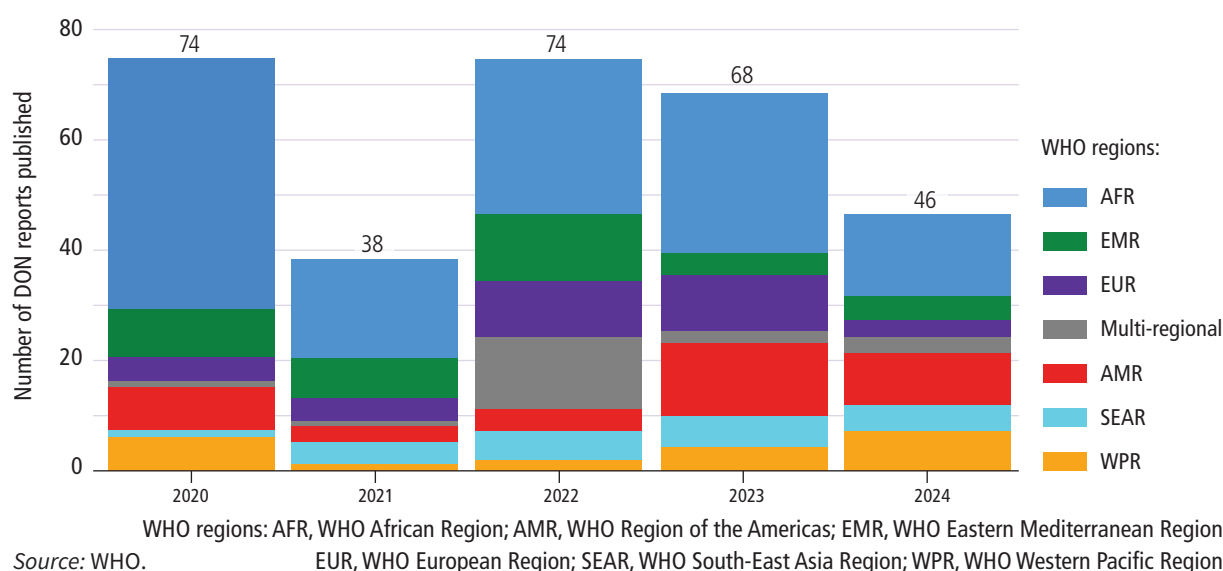
Source: WHO.

MERS, Middle East respiratory syndrome coronavirus

DON reports

Between 2020 and 2024, a total of 300 DON reports were published, averaging 60 reports annually, with a range of 38 to 74 (Fig. 18). The annual publication of DON reports showed variability over the

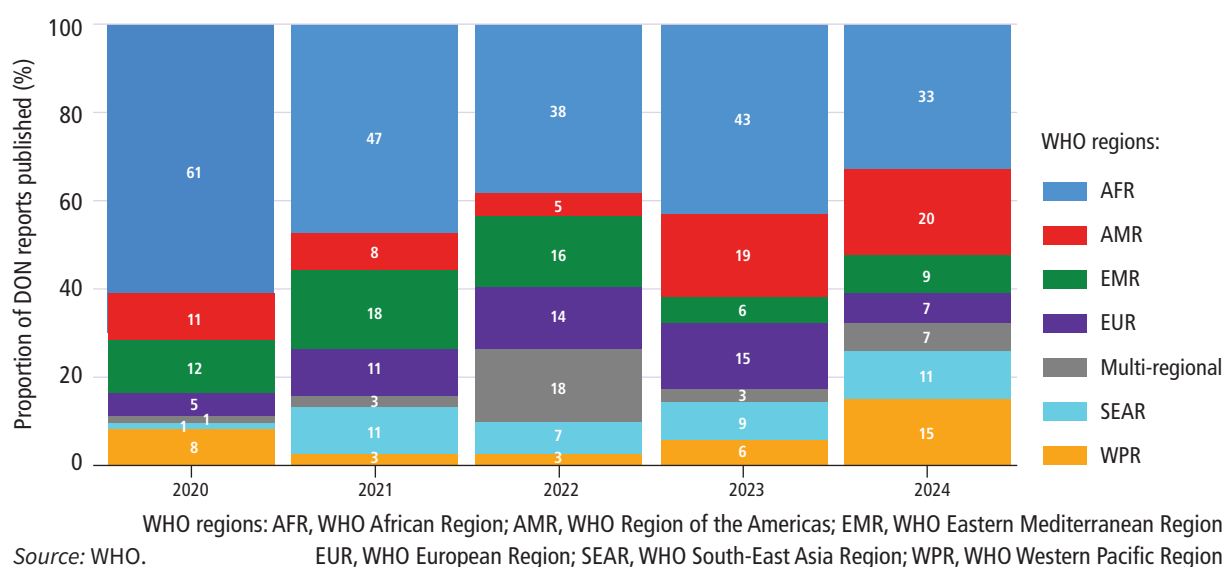
past five years, notably experiencing a decline in 2021, the second year of the COVID-19 pandemic. The subsequent rise in 2022 was primarily attributed to reports concerning mpox and cholera.

FIG. 18 Number of DON reports published by WHO region, 2020–2024

The reduction in DON reports noted in 2024 is associated with the transition from DON reports to situation reports for ongoing emergencies related to mpox and cholera.

Nearly half of DON reports published between 2020 and 2024 were related to acute public health events in the WHO African Region (45%, 135/300), followed by reports for events in the WHO Eastern Mediterranean Region (12%, 36/300) (Fig. 19). A comparable percentage of DON reports were published for both the WHO Region of the Americas (12%, 37/300) and

the WHO European Region (10%, 31/300). On the other hand, fewer DON reports were published for the WHO South-East Asia Region (7%, 21/300) and events classified as "multi-regional" (7%, 20/300), which affected multiple WHO regions simultaneously. Over the past five years, the proportion of DON reports related to the WHO African Region has shown a marked decline, dropping from 61% in 2020 to 33% in 2024; while an increase is observed for the WHO Region of the Americas, where the proportion of DON reports has risen from 11% in 2020 to 20% in 2024.

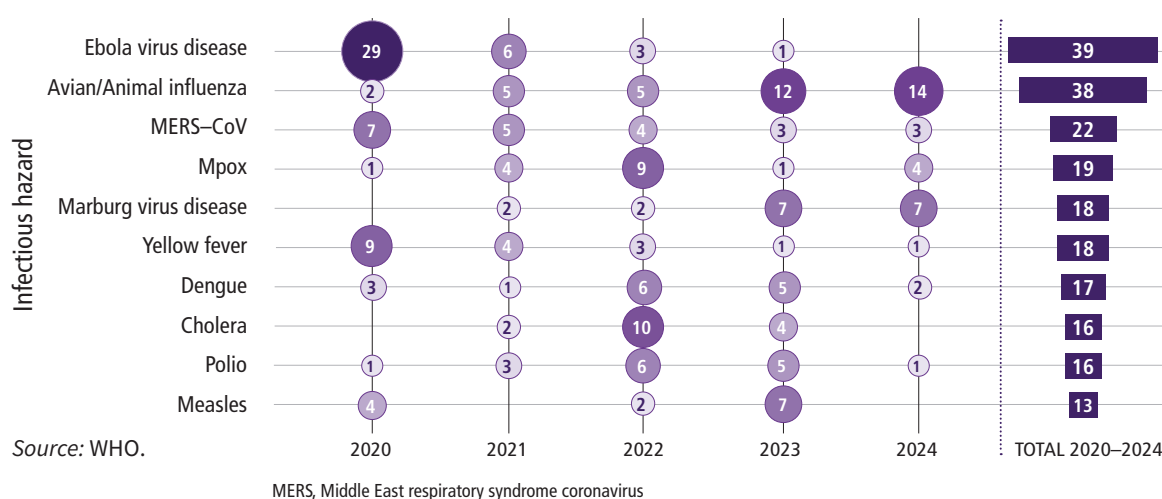
FIG. 19 Proportion of DON reports published by WHO region, 2020–2024 (in %)

All DON reports published during the five-year period were for infectious hazards.

The ten most common diseases covered by DON reports during the period 2020–2024 are shown below (Fig. 20). Ebola virus disease have the highest number of DON reports over the five-year period, with 39, followed by avian/animal influenza with 38, though their distribution across years varies significantly, with avian/animal influenza showing more consistent and

increasing reporting. MERS-CoV follows as the third most reported disease with 22 DON reports, demonstrating a notable presence, particularly in 2020. Mpox and Marburg virus disease have 19 and 18 reports, respectively, with cholera and mpox showing distinctive spikes in 2022. Figure D in Annex 2 provides a detailed analysis of the five most frequently reported diseases, conditions or hazards of an infectious nature published in DON reports by WHO region, across the five-year period 2020–2024.

FIG. 20 DON reports published for the 10 most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024



Emergency situation reports

For events that have been determined to be a Public Health Emergency of International Concern (PHEIC), events that involve multiple WHO regions or events that necessitate frequent updates due to a rapidly evolving or protracted situation, the WHO headquarters PHI team coordinates the issuance of dedicated emergency situation reports.

During the period 2020–2024, a total of 467 emergency situation reports were published, with an average of 93 reports per year (Fig. 21). The overall trend shows a decrease from 229 emergency situation reports published in 2020

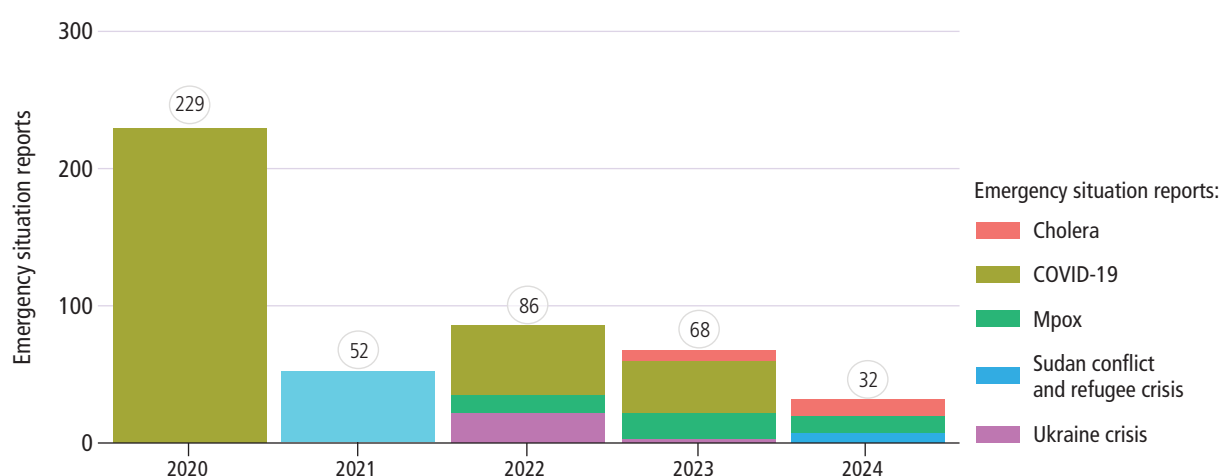
to 32 in 2024, reflecting the COVID-19 pandemic trend. The elevated number of reports recorded in 2020 was a result of the daily issuance of COVID-19 emergency situation reports. In August 2020, the reporting frequency shifted to weekly and later transitioned to monthly in September 2023, aligning with WHO move from emergency response to long-term disease prevention and control.

Over the past five years, the majority of the emergency situation reports were related to COVID-19 (79%; 370), followed by mpox (9%;

44), the Ukraine emergency (6%; 25), cholera (4%; 21), and the Sudan conflict and refugee crisis (2%; 7). In 2024, mpox has become the emergency most frequently covered by emergency situation reports (41%; 13), closely followed by cholera with 12 (37%). The Sudan conflict and refugee crisis accounts for seven reports (22%).

In addition to emergency situation reports, WHO continues to share regularly updated epidemiological data on COVID-19, cholera and mpox in the following respective dashboards: WHO COVID-19; Global Cholera Cases, and Deaths and Global Mpox Trends.

FIG. 21 Number of emergency situation reports prepared by WHO HQ PHI, by type and year, 2020–2024*



* Please note that this is not reflective of all situation reports developed by WHO.

Source: WHO.

2.5 Timeliness of Public Health Intelligence activities

Timeliness detect and respond indicator

Timeliness is a critical aspect of improving public health impact in order to protect lives. It is a reflection of the speed at which public health events are identified and responded to. GPW13, defines the detect and respond indicator, which focuses on three key aspects of timeliness: time to detect (t0), time to notify (t1), time to respond (t2). WHO leverages the 7–1–7 target, adapted to IHR notifiable events only, for detection, notification and response using the following three timeliness metrics:

- ≤ 7 days to detect a suspected public health threat;

- ≤ 1 day to notify WHO under IHR; and
- ≤ 7 days to complete early response actions.

During the past five years (2020–2024), there were 375 events reported across all WHO regions that had an EIS bulletin published and had sufficient information available to calculate the detect and respond indicator's three components.

Over this period, the median "time to detect" events was 7 days (IQR 2–19), the median "time to notify" events to WHO was 2 days (IQR

0.75–9.25), and the median "time to respond" (from the date the event was detected) was 0 days (IQR 0–4 days) (Fig. 22). The yearly analysis of the detect and respond indicator (Fig. 23) shows that in 2020, the median detection time was 1 day. From 2021 onward, it increased, reaching 29 days in 2023. In 2024, the median detection time dropped to 10.5 days (IQR = 6 – 26.75 days), but still higher than the recommended 7 days. The median notification time increased from around 1 day in 2020 to 6 days in 2021, before falling to 2 days in 2022. In 2023, it was again low at 1 day before rising to approximately 4 days in 2024. Notably, 2021 showed a broader spread of

notification times compared to the other years. Finally, the longest response time was observed in 2024, where the median response time reached 6 days, while the previous four years had lower medians. This substantial increase in time to notification in 2024 could be due to improved data collection, resulting in more data points to be analysed.

Figures in Annexes F and G show the timeliness analyses for detection, notification and response, by WHO region, for each year of the past five years, and for the combined period 2020–2024.

FIG. 22 Timeliness of events published in the EIS for detection, notification and response, 2020–2024 (in days)

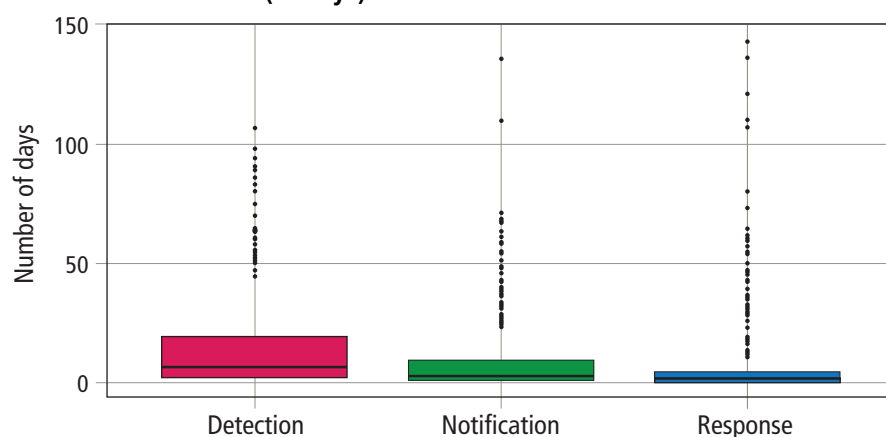
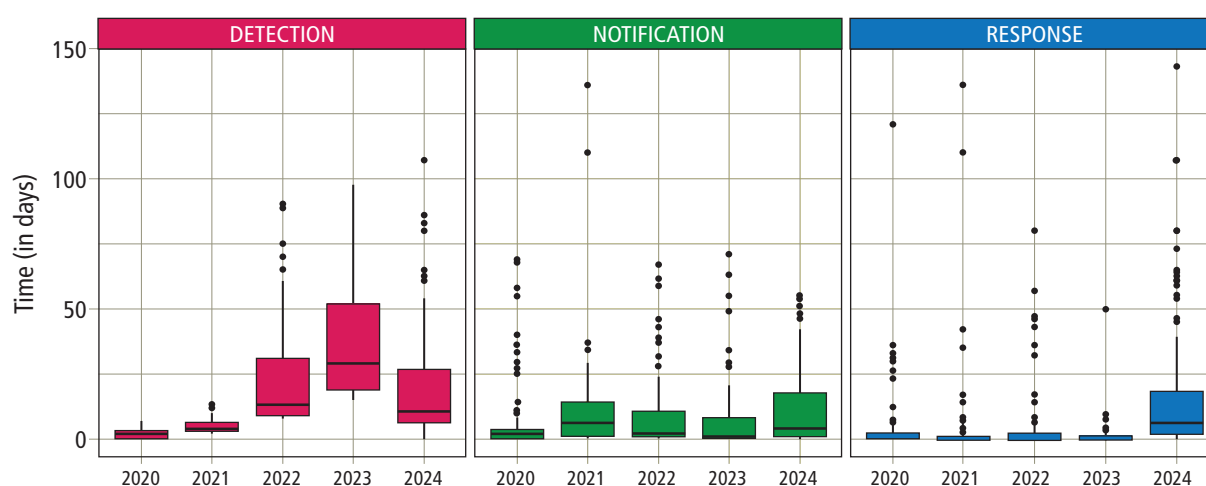


FIG. 23 Timeliness of events published in the EIS for detection, notification and response, in days, for each year of the past five years, 2020–2024



Timeliness of information products

Information dissemination is key to an effective rapid response to acute public health events. Under the IHR mandate, WHO produces two key information products as part of its public health intelligence activities (as described in section 2.4): postings on the EIS for NFPs and publishing DON reports for the public. Some events also undergo a more in-depth RRA conducted independently by WHO.

These three information products, mandated under the IHR, are coordinated by the PHI Unit at WHO HQ, which has established the following timelines from initiation to dissemination:

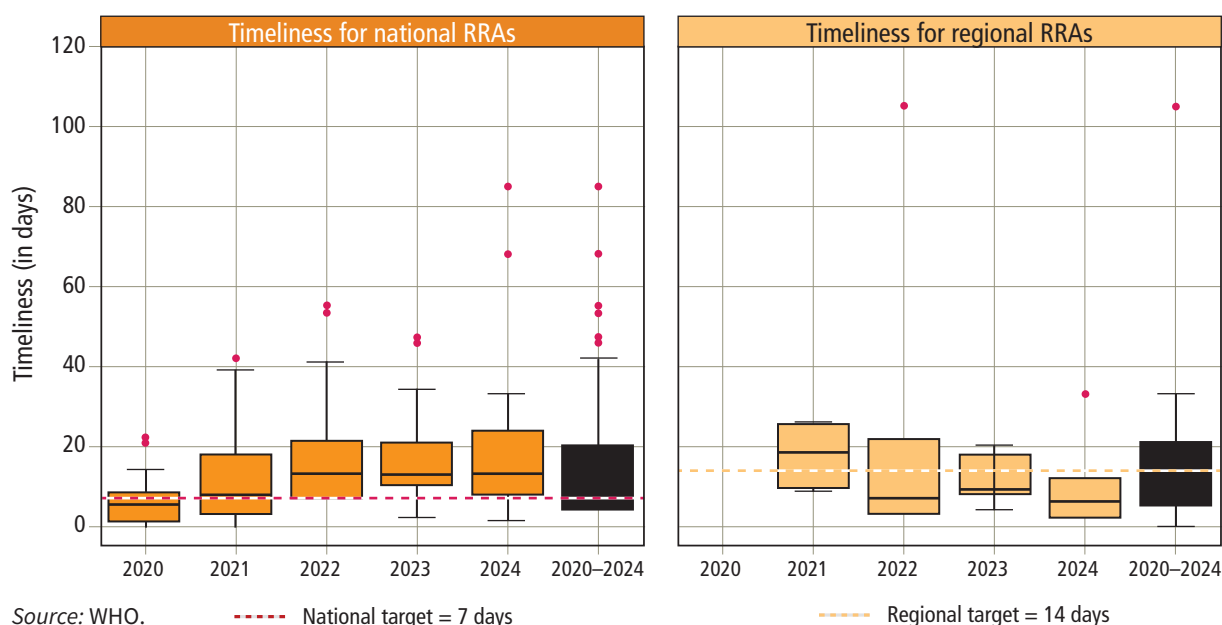
- **RRA reports:** 7 days for national RRAs, 14 days for regional RRAs;
- **EIS postings:** 4 days; and
- **DON reports:** 5 days.

Timeliness of RRAs

Out of the 195 national and regional RRA reports published between 2020 and 2024 (Fig. 24), 192 (98%) had information available to calculate the timeliness indicator; two national and one regional did not have the data available. Between 2020–2024, national RRAs (173) had a median of 10 days from initiation to dissemination, above the 7-day target, with 38.1% (66/173) being completed within this 7-day target.

The median increased from 5.5 days in 2020 to 13 days in 2024, with the number achieving the timeliness target dropping from 68.4% (26/38) to 26.6% (4/15) over the same period. Regional RRAs (19) have a median of 9 days (2021–2024), with 63.1% (12/19) being completed within the 14-day target; no regional RRAs were published in 2020. In 2021, the median exceeded the target, reaching 18.5 days.

FIG. 24 Timeliness of national and regional RRAs by year and by period, 2020–2024

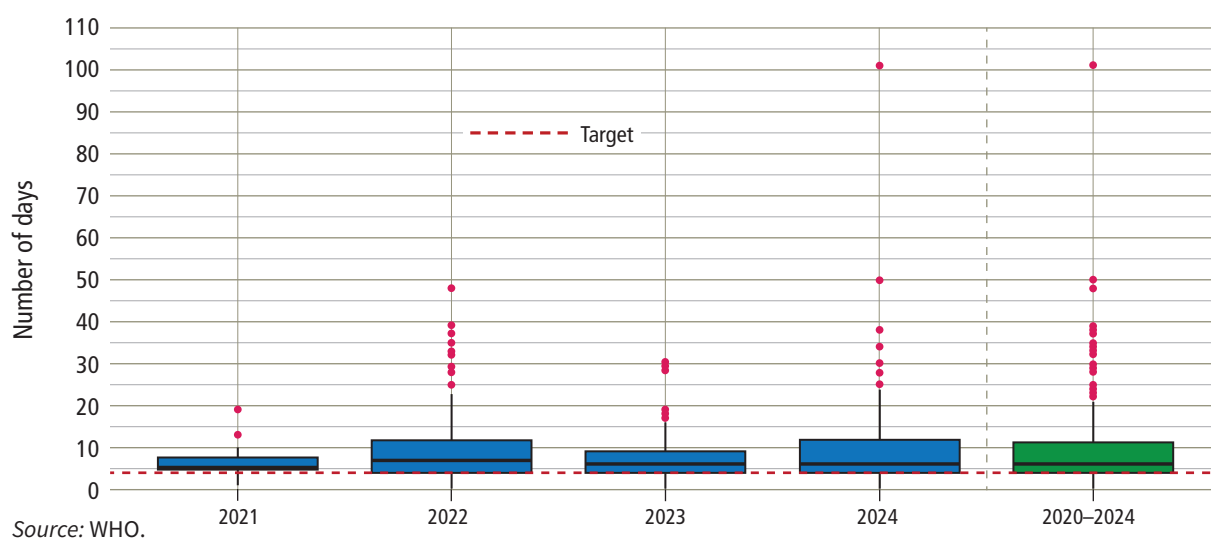


Timeliness of EIS bulletins

Data for calculating the timeliness indicator for EIS bulletins have been available since 2021 for a total of 338 EIS bulletins [90%(338/375) of total EIS bulletins published between 2021 and 2024]. The aggregated 2021–2024 analysis shows that the overall median time from EIS initiation to dissemination is 6.0 days, with an average of 9.1 days (Fig. 25). This corresponds to a compliance rate of 30.2% against the 4-day target.

In 2024, the median time was 6.0 days with an average of 9.8 days, and a compliance of 31.7%. Meanwhile, in 2023 the median remained at 6.0 days with the mean at 7.7 days and a compliance rate of 30.9%. These findings indicate relatively consistent performance over the years, although the processing times are above the desired target.

FIG. 25 Timeliness of EIS bulletins published by year and aggregated across the period 2021–2024

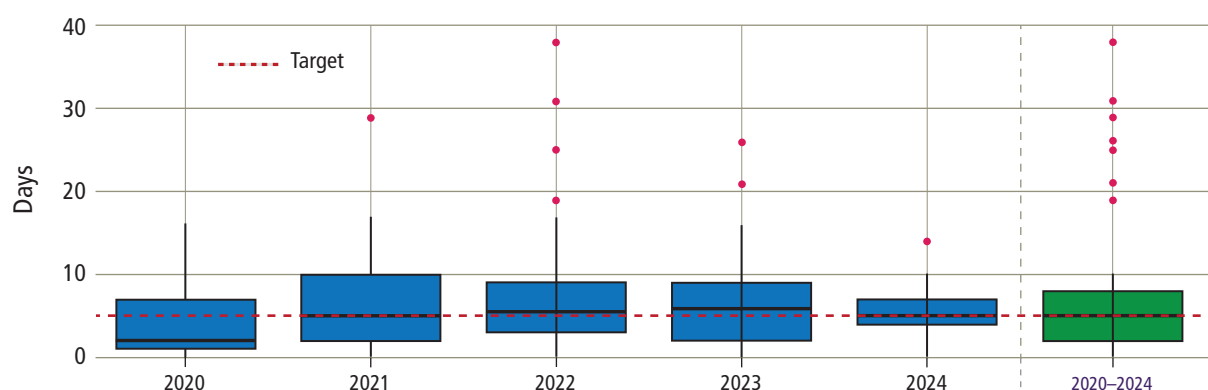


Timeliness of DON reports

All the 300 DON reports issued between 2020 and 2024 had information to calculate the timeliness indicator. In the aggregated 2020–2024 period, the median time from DON initiation to share in 5 days, which exactly meets the established target (Fig. 26). However, compliance analysis shows that approximately 50% of reports exceed this target, indicating important variability. In 2020, the median time was 1.5 days, the lowest value of

the entire period, with a compliance of approximately 66.7%. Subsequently, the median time progressively increases: in 2022 and 2023, median values exceed the target, reaching approximately 7 days, with compliance dropping to around 40%.

Finally, in 2024, the median time decreased to 4 days with a compliance rate of 55%, indicating an improving trend.

FIG. 26 Timeliness of DON reports published by year and aggregated across the period 2020–2024

Source: WHO.

2.6 Public Health Intelligence in countries requiring international humanitarian assistance

PHI is just as crucial in humanitarian settings as it is in other emergency and non-emergency contexts. Humanitarian crises often create conditions that heighten epidemic risks, disrupting key public health measures like vaccination and vector control, displacing populations, limiting access to safe water and sanitation, and creating barriers to healthcare. Additionally, large-scale epidemics in these settings can further strain health systems by redirecting critical resources toward outbreak response. Therefore, a robust early warning, alert, detection and response (EWAR) system is essential in humanitarian contexts to enable the timely identification and containment of epidemics.

In 2024, among the 39 countries with a Humanitarian Response Plan (HRP), 27 (69%) were affected by newly reported outbreaks, followed by 13 countries (33%) affected by new natural disasters, 7 (18%) by societal hazards and 1 (3%)

by product hazards (i.e. counterfeit medicine) (Fig. 27).

Communication on events in humanitarian settings is often included in the respective country's situation reports or health cluster bulletins; however, information dissemination under IHR does still occur. In 2024, among the 39 countries with a HRP, 30 (76.9%) recorded a new event in EMS, with Guinea-Bissau reporting the highest number (7 events; 8.8%). A total of 10 EIS bulletins were published across 8 HRP countries, with Nigeria and Timor-Leste each publishing 2 EIS bulletins (20% each). Among the 10 RRAs published across 6 HRP countries, half (5 RRAs; 50%) were related to Nigeria. Additionally, 6 DONs were published across 4 HRP countries, with the Democratic Republic of the Congo accounting for half of them (3 DONs; 50%) (Fig. 28).

FIG. 27 Countries, territories or areas with a Humanitarian Response Plan by type of hazard, 2024

Country, territory or area	TYPE OF HAZARD			
	INFECTIOUS	DISASTER	SOCIETAL	PRODUCT
Afghanistan	2			
Burkina Faso	2			
Burundi	3	1		
Cameroon	1			
Central African Republic	2		1	
Chad	1		1	
Comoros	1			
Democratic Republic of the Congo	3	1		
Congo	1	1		
Eritrea				
Ethiopia		3	1	
Guinea-Bissau	5	2		
Haiti	2			
Iraq				
Kiribati	1			
Lebanon	1		1	
Libya		1	1	
Mali	1			
Marshall Islands				
Micronesia (Federated States of)				
Mozambique	2			
Myanmar	1	1		
Niger	5	1		
Nigeria	4	1		
occupied Palestinian territory, including east Jerusalem				
Papua New Guinea		2		
Sao Tome and Principe	1			
Solomon Islands				
Somalia	1			
South Sudan	4	1	1	
Sudan				
Syrian Arab Republic	2		1	
Timor-Leste	2			
Tuvalu				
Ukraine	1			
Venezuela (Bolivarian Republic of)	2			1
Yemen	2	1		
Zimbabwe	2	1		
Kosovo*				

* All references to *Kosovo* in this document should be understood to be in the context of the United Nations Security Council resolution 1244 (1999).

Source: WHO.

FIG. 28 Key Public Health Intelligence products (EMS events, EIS bulletins, RRA and DON reports) by country, territory or area with a Humanitarian Response Plan in 2024

Country, territory or area	New events recorded in EMS	EIS bulletins published	RRA reports published	DON reports published
Afghanistan	2			
Burkina Faso	2			
Burundi	4			
Cameroon	1			
Central African Republic	3			
Chad	2	1	1	1
Comoros	1			
Democratic Republic of the Congo	4	1		3
Congo	2			
Eritrea				
Ethiopia	4	1	1	1
Guinea-Bissau	7	1		
Haiti	2			
Iraq				
Kiribati	1			
Lebanon	2			
Libya	2			
Mali	1			
Marshall Islands				
Micronesia (Federated States of)				
Mozambique	2			
Myanmar	2		1	
Niger	6			
Nigeria	5	2	5	
occupied Palestinian territory, including east Jerusalem				
Papua New Guinea	2			
Sao Tome and Principe	1			
Solomon Islands				
Somalia	1	1	1	
South Sudan	6	1		
Sudan				
Syrian Arab Republic	3			
Timor-Leste	2	2	1	1
Tuvalu				
Ukraine	1			
Venezuela (Bolivarian Republic of)	3			1
Yemen	3			
Zimbabwe	3			
Kosovo*				

* All references to *Kosovo* in this document should be understood to be in the context of the United Nations Security Council resolution 1244 (1999).

Source: WHO.

Public Health Intelligence during mass gatherings

WHO defines mass gathering as a planned or spontaneous event where the number of people attending the event could strain the planning and response capacity of the community or country hosting the event. These events can include international sporting tournaments such as the Olympic Games, religious pilgrimages like the Hajj, political rallies, cultural events and large conferences.

Recognizing the public health challenges posed by mass gatherings, WHO provides evidence-based guidance, technical support, and strategic advice to host governments, health authorities and event organizers. The goal is to strengthen preparedness, minimize health risks, and maximize the long-term positive health legacy of these events.

WHO plays a key role in providing technical guidance and support to Member States and event organizers on mass gathering events. However, WHO does not have the mandate to enforce any action with regard to a mass gathering (modification, postponement, cancellation, etc.) or to authorize that it may proceed.

WHO's work on mass gatherings for health preparedness is rooted in key international public health frameworks, including the IHR, which mandate that WHO works with Member

States to develop policies, measures and tools for managing the potential health risks associated with large events. Additionally, a 2011 decision by the WHO Executive Board emphasized the need for close collaboration between WHO and Member States to support intersectoral cooperation in the planning and execution of mass gatherings.

In the context of mass gatherings, event-based surveillance (EBS) plays a critical role in early detection of public health threats, not only in the host country, but also in the home countries of attendees and neighbouring nations. Enhanced EBS has been recognized as a crucial component of mass gathering health preparedness since its emphasis in a WHO Executive Board report published in 2011 (*Global mass gatherings: implications and opportunities for global health security*). It has remained a priority in WHO guidance.

In 2024, WHO provided technical support for enhanced public health intelligence activities at a number of mass gathering international events, including the Hajj, the UEFA European Championship, and the Paris 2024 Olympic and Paralympic Games. These efforts helped to monitor disease transmission, strengthen emergency response capacities and ensure safer gatherings through coordinated global action.

Hajj

The Hajj in the Kingdom of Saudi Arabia (KSA) stands as the world's largest annual religious mass gathering, attracting around two million Muslim pilgrims to Mecca for Hajj rituals. KSA has extensive experience in managing such gatherings and maintains a robust surveillance system capable of swiftly detecting potential outbreaks during Hajj.

The 2024 Hajj took place from 14–19 June. The WHO Regional Office for the Eastern Mediterranean conducted its enhanced event-based surveillance (EBS) activities from 7 to 23 June, which allowed a buffer of 7 days prior and 3 days after the event. The monitoring was carried out in collaboration with the Gulf Centre for Disease Prevention and Control (GCDC) and the European Centres for Disease Prevention and Control (ECDC). Prior to the event, a standard operating procedure (SOP) was drafted based on the SOP of the Eastern Mediterranean Region's PHI team and shared with partners (GCDC and ECDC). The SOP included the high priority health

hazards during Hajj (communicable, noncommunicable and environmental hazards like heat stroke, trauma, road accidents, etc.). EBS was conducted using the specific Hajj monitoring board on the Epidemic Intelligence from Open Sources (EIOS). This was complemented by the in-house social media targeted search and the routine screening. In addition, GCDC and ECDC shared their respective reports to the PHI team of the Eastern Mediterranean Region on daily basis. Duplication of reports is screened out during the daily triage process. Reports are prepared and shared with the WHO Country Office in KSA for approval by the WHO Representative, prior to sharing with the NFP of KSA.

From 7 to 23 June 2024, there were 13 204 articles screened from EIOS, 11 signals were classified as “for information only” and no signal warranted verification. The most frequent signal captured during the Hajj period was the number of people who suffered or died from heat stroke.

UEFA Champions League

Germany was chosen as the host country for the Union of European Football Associations (UEFA) 2024 Championship, more commonly known as UEFA EURO 2024 or the “Euros”. The matches were held from 14 June to 14 July 2024, attracting 2.68 million attendees/fans from more than 190 countries, holding games across 10 cities and stadiums throughout Germany.

A collaboration between the WHO Regional Office for Europe, ECDC and the Robert Koch Institute

(RKI) was established for the championship. Each organization conducted EBS, with a focus on media monitoring, through their individual boards in the EIOS platform. In order to remove duplication of efforts, each organization had a different geographical and threat focus, with RKI focusing on biological threats within Germany, ECDC focusing on biological threats within the European Union/European Economic Area (EU/EEA), and with the WHO Regional Office for Europe focusing on biological threats in non-EU/

EEA countries and non-biological threats within the WHO European Region through two boards created in the EIOS platform.

Priority diseases and hazards were identified based on the WHO Regional Office for Europe's routine media monitoring objectives, which are rooted in the IHR. This surveillance effort began on 1 June 2024, two weeks before the start of the event, and was completed two weeks after the event on

28 July 2024. A dashboard was created for the event that incorporated signals from all three organizations. Across all countries, 29 signals were detected. Among these detected signals, 79.3% were associated with non-biological hazards and 20.7% were associated with biological hazards. While several signals with potential public health impact on the Games were flagged, none were assessed as requiring notification through the IHR mechanism.

Paris Olympic and Paralympic Games

Paris was chosen as the host city for the 2024 Summer Olympics, held from 26 July to 11 August, with the Paris 2024 Paralympic Games taking place from 28 August to 8 September. A total of 14 900 athletes participated in the games, including 10 500 for the Olympic games and 4400 for the Paralympic games. The games attracted around 12 million visitors, including 9.5 million for Olympics games and 2.5 million for Paralympic Games. Priority diseases and hazards were identified based on WHO's prior EBS activities for other Olympic Games, the French government's priority disease list, and hazards flagged by the WHO European Regional Office for the UEFA Champion League 2024, a list guided by risk assessment to ensure the surveillance system is focused on the most relevant and high-impact health threats specific to the event and its context. Enhanced EBS was conducted using two media monitoring boards developed on the EIOS platform, covering both domestic and global health signals.

This surveillance effort spanned from two weeks before the Olympic Games to two weeks after

the Paralympic Games. Signals were assessed for their potential risk to the Games and the general public, and findings were shared with WHO's Mass Gathering experts, ECDC and the WHO European Regional Office, and the WHO Western Pacific Region (for French Polynesia) daily. The EBS activities were complemented by routine surveillance and collaboration with other entities' EBS initiatives for an enhanced PHI approach.

Between 12 July and 20 September 2024, a total of 42 870 raw signals were scanned and triangulated, with 49 (0.1%) signals assessed and triaged for potential public health risks. Of these, 16 were further assessed to have a potential impact on the Games, with respiratory diseases, particularly COVID-19, accounting for 44% ($n = 7$), followed by soil- and water-borne diseases (25%, $n = 4$). While several signals with potential public health impact on the Games were flagged, none were assessed as requiring notification through the IHR mechanism.

ICC Men's T20 World Cup (West Indies & United States of America, 2024)

The International Cricket Council (ICC) biennial world cup, the ICC Men's T20 World Cup, was co-hosted by the United States of America and the West Indies, and took place in three stadiums in the United States of America (in the states of Texas, Florida and New York), as well as in six venues in the West Indies (in Barbados, Saint Lucia, Saint Vincent, Trinidad and Antigua) with the event featuring 20 teams competing across 55 games from 1 to 29 June 2024. The monitoring was carried out between 15 May to 15 July 2024 (15 days before the opening to 15 days following the closure of the competition) and parameters were set for selecting pieces of information using publicly available information with an all-hazard approach, in order to rapidly detect public health risks which can be caused by the circulation or gathering of people during the tournament.

The WHO Region of the Americas developed a Media Monitoring Dashboard, exclusively accessible through a password to authorized countries and territories in the Region involved in the Cup. Users were able to view daily health-related updates for each participating country, filter content by date, country, source type (official and unofficial), hazard type and export daily reports compiling key information. Additionally, the dashboard allowed access to past records of all these selected items. By including data from both official national reports and a variety of unofficial media sources, the dashboard aimed to offer comprehensive coverage. During the event, it received 456 views from 25 countries in the Region. This initiative underscored the value of a secure, centralized tool that provides local managers and event organizers with timely access to health and safety information, enabling them to quickly address potential health risks impacting host locations during large-scale events.

Public Health Intelligence in the One Health Framework: GLEWS+

Emerging and re-emerging zoonotic diseases pose a constant threat to global health security, driven by climate change, environmental degradation, and human and animal movement. Delays in detecting outbreaks and information gaps can lead to widespread outbreaks, economic disruptions, and increased morbidity and mortality in both humans and animals.

Recognizing the need for a coordinated approach to early warning and response, the Global Early Warning System (GLEWS) was launched in 2006 as a joint initiative between FAO, WHO and WOA, known collectively as the Tripartite. Initially, it focused on monitoring and sharing information on zoonotic and transboundary animal diseases, using animal outbreaks as early warnings for public health surveillance. Over time, the growing complexity of health threats highlighted the need for an enhanced, more integrated approach to public health intelligence.

In response, GLEWS+ was developed to expand the system's scope beyond zoonotic disease tracking, to include diseases of unknown etiology, incorporating broader event-based and indicator-based surveillance mechanisms, enhanced risk assessment, and cross-sectoral intelligence-sharing at the human-animal-ecosystem interface.

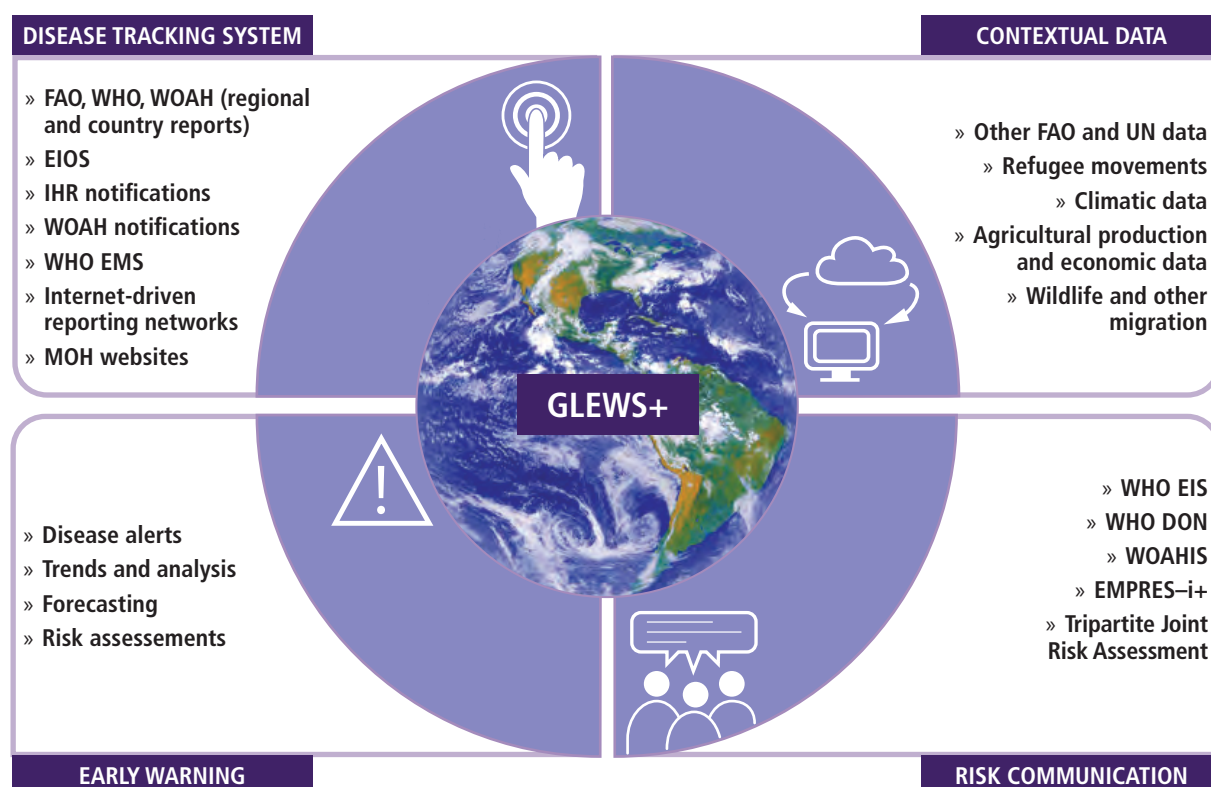
This evolution strengthened its role as a One Health intelligence platform, improving detection, coordination, and risk assessments to inform global response efforts.

The PHI Unit at WHO HQ is tasked with detecting and sharing early warning signals of potential health threats with partners across the Tripartite. These signals are then verified and assessed through collaborative efforts. Once health events are confirmed, they are shared with decision-makers (e.g. Ministries of Health), allowing for timely risk assessments and coordinated response efforts at national, regional and global levels.

For zoonotic health threats and diseases of unknown etiology, PHI works with GLEWS+, a core One Health intelligence network shared across the Tripartite.

GLEWS+ is not a separate initiative, but an integral part of PHI's operational workflow. Within this framework, the WHO headquarters' PHI team leverages surveillance data, technical expertise, and risk assessments to detect and assess disease threats at the human-animal-ecosystem interface, enabling PHI to efficiently monitor and respond to these specific events (Fig. 29).

FIG. 29 Overview of the One Health, Global Early Warning System for health threats and emerging risks at the human–animal–ecosystem interface (GLEWS+)



DON, Disease Outbreak News; **EIS**, Event Information Site (WHO); **EIOS**, Epidemic Intelligence from Open Sources; **EMPRES-i+**, Global Animal Disease Information System (FAO); **EMS**, Event Management System; **FAO**, Food and Agriculture Organization of the United Nations; **IHR**, International Health Regulations; **MOH**, Ministry of Health; **UN**, United Nations; **WHO**, World Health Organization; **WHOAHIS**, World Animal Health Information System; **WOAH**, World Organisation for Animal Health

Source: GLEWS+, the Joint FAO–WHO–WOAH Global Early Warning System for health threats at the human–animal–ecosystem interface. All rights reserved

The WHO headquarters' PHI team continuously monitors a range of official and unofficial sources through GLEWS+, including:

- ▶ Official reports from public health and animal health authorities at a global level;
- ▶ Surveillance networks and early warning mechanisms of FAO, WHO and WOAH;
- ▶ Open-source epidemic intelligence platforms, such as WHO's Epidemic Intelligence from Open Sources (EIOS); and
- ▶ Media reports and alerts from trusted partners.

The WHO headquarters' PHI GLEWS+ focal points evaluate identified signals against established criteria, including a defined list of priority diseases and their potential impact on public and animal health. Signals meeting these criteria are documented and tracked within the secure, web-based GLEWS+ platform, which serves as a confidential central database. This platform supports trend analysis, secure information exchange and timely communication across the Tripartite.

Early warning notifications generated via GLEWS+ are disseminated through each organization's respective early warning mechanisms for further verification and additional data collection. Confirmed disease events often prompt formal Tripartite joint risk assessments (JRAs), conducted by technical experts from FAO, WHO and WOAH. GLEWS+ additionally provides essential information and analysis contributing directly to WHO Risk Assessments and Event Information Sheets (EIS) for zoonotic and emerging health threats.

Through joint recommendations, collaborative risk assessments, and technical support facilitated by GLEWS+, PHI helps guide public health interventions and outbreak containment strategies, including:

- Risk communication and recommendations for interventions across human and animal sectors.
- Collaboration between human health and animal health authorities to coordinate response actions.

The trend for confirmed events recorded and followed in the GLEWS+ platform from 2020 to 2024 is shown below (Fig. 30). Over the past five years, a total of 277 events have been recorded on the GLEWS+ platform, with an average of 55 events per year. During this period, the number of events gradually increased reaching a peak in 2023 with 95 events. This sharp increase in health events can be attributed to several factors, including, an increased capacity in the GLEWS+ tripartite team and the extension of the network to WHO regional offices. Also, it is possible that more events occurred globally during this period, due to the rapid rate of urbanization and travel, increased animal and population movement, climate change, virus evolution and adaptation.

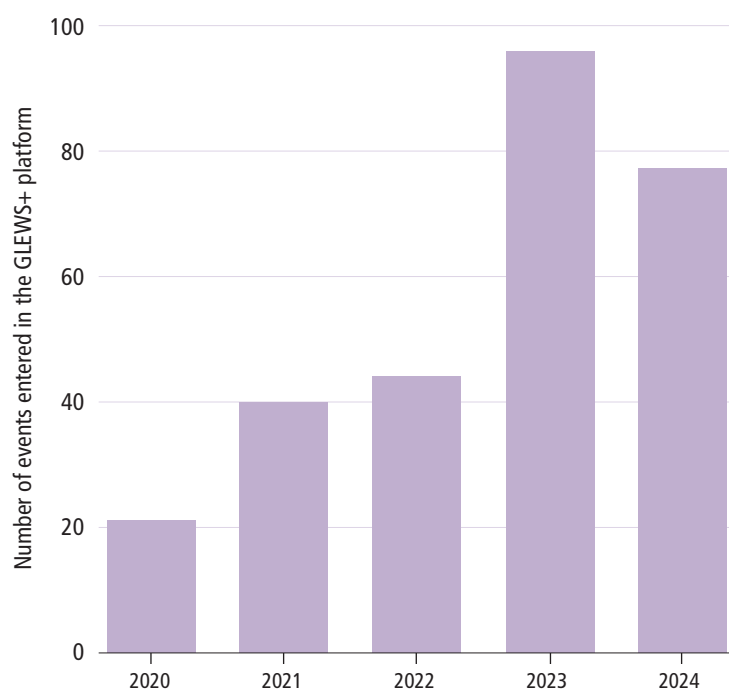
Additionally, improvements in surveillance systems, increased public awareness, and better reporting capabilities also contributed to the higher number of reported events.

In 2024, a 19% decrease in the number of recorded events was observed. Of the 79 confirmed events, majority of events, 30% were related to highly pathogenic avian influenza A (H5) human infections, followed by 6.3% events of Rift Valley fever and anthrax, 3.8% of equine encephalitis, and an equal 2.5 % of cases of New World screwworm, Ebola virus disease, Marburg virus disease and Nipah virus disease.

As zoonotic and emerging disease threats continue to evolve, GLEWS+ remains essential to global health security. Integrated within the WHO headquarters' PHI operations, the system supports early detection, rapid response, and effective cross-sectoral coordination.

Through collaborative surveillance and intelligence sharing, GLEWS+ has significantly enhanced global capacities, enabling the timely identification of high-impact outbreaks such as avian influenza (H5N1) or Ebola. By fostering greater transparency and trust in disease reporting, it has strengthened cooperation among countries and international partners, ultimately leading to faster and more informed decisions.

Looking forward, GLEWS+ will continue to play a critical role in global preparedness, risk assessment, and response, reinforcing WHO's commitment to protecting human, animal and environmental health.

FIG. 30 Annual trend in the number of events entered in the GLEWS+ platform, 2020–2024

Source: WHO.

Public Health Intelligence in action

5.1 Armed conflict and refugee influx in the countries of the WHO Eastern Mediterranean Region

The WHO Eastern Mediterranean Region has been facing several challenges over the years. Out of the 22 countries and territory, nine are classified as fragile, conflict and vulnerable, with more than 100 million people in need of humanitarian assistance, which represent 36% and 15% of the regional and global population, respectively. The Region also accounts for 58% of the global refugee burden. In 2024, the Regional Office has been monitoring three armed conflicts with humanitarian and health consequences.

Lebanon hostilities: since 7 October 2023, Lebanon has experienced a prolonged conflict that resulted in significant human casualties and extensive material damage. As of 2 December 2024, there were 3961 deaths and 16 520 injuries. Despite the ceasefire agreement, which came into effect on 27 November 2024, humanitarian needs remain at unprecedented levels. On 25 November 2024, the number of internally displaced persons peaked at 899 725. As such, the Public Health Situational Analysis (PHSA) has been updated and Lebanon hostilities has been graded as Grade 3, thereby activating the internal WHO ERF. The Incident Management Structure (IMST) has been established, where PHI has been providing updates on the signals as well as taking part in updating the PHSA.

Syrian Arab Republic escalation of hostilities: protracted hostilities, repeated displacement of civilians, economic recession, disease outbreaks and a devastating earthquake have combined to create one of the world's most

complex emergencies for Syrians. On the early morning of 27 November 2024, a large-scale offensive was launched by allied non-state armed groups from Northwest Syria (NWS) against areas controlled by government forces. There were reports of shelling, airstrikes and other attacks between 26 and 28 November, affecting 50 communities across three governorates. Casualties were reported on both sides, with estimates exceeding 100 combatants, alongside civilian casualties and injuries. This escalation is unprecedented in scale and seeks to shift control over territories held by government forces since the last major offensive in 2020. The PHSA was immediately developed across the three levels, graded the emergency as Grade 3 and IMST was established. Regular updates are being done across pillars through this platform.

Yemen armed conflict: after 10 years of conflict, Yemen continues to suffer severe deterioration of economic conditions, extensive damage to civilian infrastructure, and the collapse of essential services. The number of people facing food insecurity peaked at 17.6 million, and nearly half of all under-five children suffer from moderate to severe stunting; 4.5 million people remain displaced, many of whom have experienced multiple displacements over several years. On 26 December, Israel Defense Force (IDF) launched airstrikes on Sana'a Airport, Hudaydah Seaport, Ras Issa Seaport and Ras Katheeb Power Station in Hudaydah Governorate. These strikes resulted in six deaths and 39 injuries, including a UN plane crew member. Continued IDF airstrikes further

worsened civilian displacement and infrastructure damage. Conflict-affected areas such as Hudaydah, Hajjah and Sa'ada are not only targets of airstrikes, but also face severe public health crises, with Hudaydah bearing the highest burden of diseases like dengue, cholera, diphtheria and measles. These overlapping challenges further complicate the ongoing crisis and strain the already limited healthcare resources. Yemen remains a protracted-3 emergency where close monitoring is being done, including outbreaks of cholera, dengue, measles and circulating

vaccine-derived poliovirus type 2 (cVDPV2). A PHSA is in progress. In summary, PHI work goes beyond infectious diseases. In all the armed conflicts mentioned above, the team in the Regional Office has shown its key function in detecting, assessing, reporting, coordinating and collaborating across the three levels of the Organization. The Emergency Operations Center (EOC) has been a venue for all staff across the two levels to update and discuss the signals, outbreaks and other acute public health events, where PHI plays a key role.

5.2 Avian Influenza A(H5N1)

The WHO Regional Office for Europe has been monitoring the influenza A(H5N1) situation both regionally and globally, employing a multifaceted approach to ensure comprehensive surveillance at the human-animal-environmental-interface. Utilizing event-based surveillance activities, IHR NFPs, partner networks [such as the European Commission (EC), European Centre for Disease Prevention and Control (ECDC), European Food Safety Authority (EFSA)] and collaborations through the quadripartite – including FAO, United Nations Environment Programme (UNEP), WOA – the Regional Office has established a robust system for timely detection and reporting of influenza A(H5N1) cases in humans, facilitating prompt and effective public health response, as needed.

Since the detection of influenza A(H5N1) in dairy cattle in March 2024 in the United States of America (USA), existing surveillance efforts in humans have been reinforced, and additional risk-based surveillance of animals with occasional milk testing has been implemented in some of the European countries. Although there are currently no reports of influenza A(H5N1) in dairy cattle in Europe, the USA experienced a high number of avian influenza outbreaks in 2024, including a spillover events into dairy herds. Dairy herds and other mammals in Europe may also be at risk of

introduction of influenza A(H5N1) from wild birds or poultry, especially during the migratory season when H5N1 outbreaks in poultry are historically more common. If H5N1 is introduced to dairy herds, there is a risk that the virus may become present in raw milk and potentially spread through the food chain, requiring strict controls and continuous food monitoring.

In the past 10 years and as of December 2024, a few human cases of influenza A(H5N1) have been reported in the Region, and none of these are known to be associated with the ongoing outbreak in the USA. Spain reported detection of H5N1 in asymptomatic workers (two in 2022), but judged these not to be true cases, rather environmental contamination, while the United Kingdom of Great Britain and Northern Ireland (United Kingdom) confirmed one true infection in 2022. Additionally, from 2021 to 2024, five other infections in the United Kingdom were deemed contamination based on further laboratory results rather than true infections.

In summary, the coordinated approach taken by the Regional Office in monitoring, preparedness and response to influenza A(H5N1) underscores the importance of multisectoral coordination, collaboration and comprehensive surveillance and risk assessment.

The WHO Regional Office for Europe continues to implement the following measures for readiness, preparedness and response

1. **Contingency planning:** in mid-June 2024, the Regional Office activated a Coordination Cell – mirroring essential functions similar to graded emergencies (IMST) – and developed a contingency plan with various scenarios to ensure operational readiness and response
2. **Guidance and resources:** the Regional Office regularly provides Member States with technical guidance on surveillance, reporting of novel influenza viruses and resources to prepare, prevent and control influenza A(H5N1) outbreaks in animals and humans. This includes operationalizing the One Health approach, enhancing surveillance and implementing food safety measures, including milk testing and performing environmental sampling.
3. **Risk assessment coordination:** the Regional Office collaborates with partners such as ECDC, EFSA and the Quadripartite to assess the risk posed by influenza A(H5N1), ensuring a comprehensive understanding of the virus's behaviour and potential threat.
4. **Quadripartite collaboration:** regular meetings are held with FAO, UNEP and WOA for monitoring the situation and conducting joint risk assessments. The Regional Office also monitors all H5N1 notifications in animals submitted to WOA and published on their website to understand any changes or patterns in the cases.
5. **Risk communication and community engagement:** an H5N1 package with FAQs for the general public and specific guidance for farm workers to strengthen preparedness has been prepared, ensuring at-risk groups are informed and equipped to protect themselves. Social listening has been conducted to inform infodemic insights on false or misleading narratives.
6. **Weekly deep dives:** the Regional Office conducts weekly deep dives on influenza A(H5N1) to stay updated on the latest global and regional epidemiological situations, allowing for adaptive response strategies.

5.3 Oropouche virus disease

Oropouche virus (OROV) is an emerging arbovirus in the Americas, causing febrile illness with systemic symptoms. An unprecedented surge in cases was reported from late 2023 to 2024 across South and Central America, and the Caribbean, with imported cases in North America and the WHO European Region (Fig. 31). Environmental changes and human movements drive the virus's spread, with deforestation, climate variability and agriculture intensification increasing spillover risk.

From 1 January 2024 to 25 January 2025, 11 countries and one territory in the Americas reported 20 004 confirmed Oropouche cases, including

four deaths, through NFPs. Since the beginning of 2024, imported Oropouche virus cases have been reported in the United States (108 cases), Canada (3 cases) and the Cayman Islands (1 case), linked to travel to Cuba and Colombia. The WHO European Region also saw its first imported cases (30 in total) in Germany, Spain and Italy associated with travel to Cuba and Brazil.

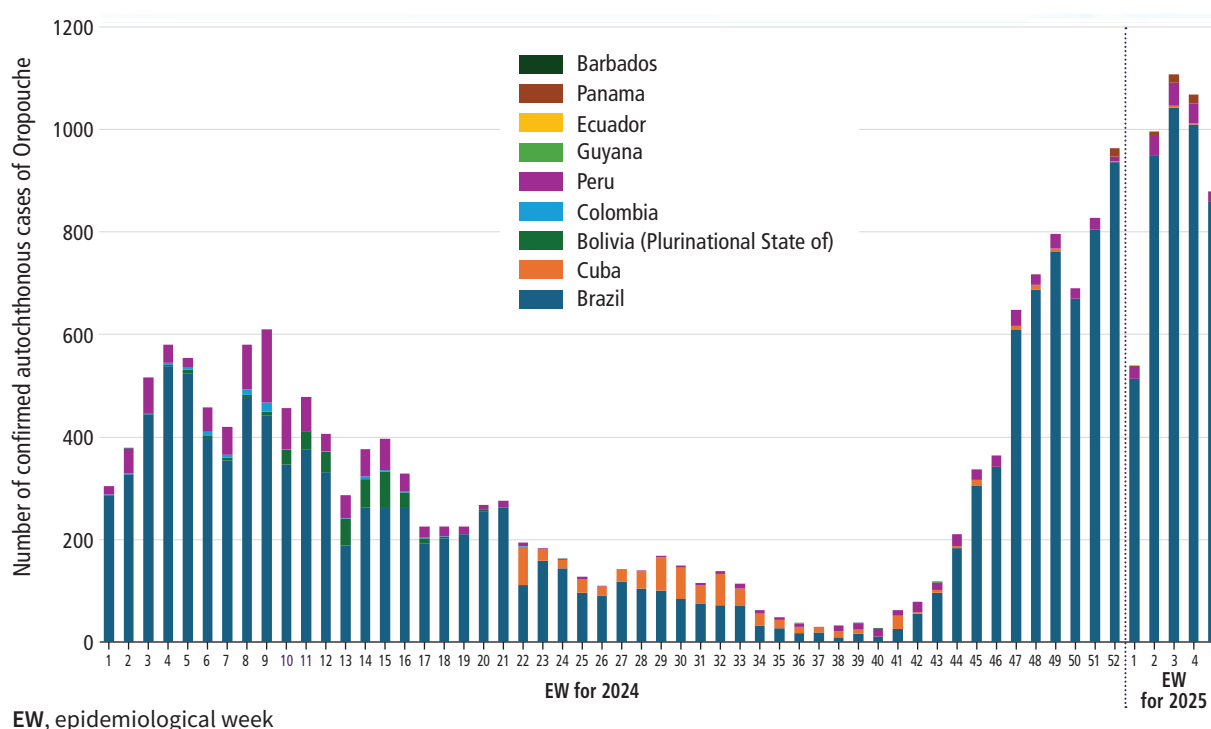
On 2 February 2024, Brazil's NFPs reported Oropouche virus outbreaks in Amazonas state, identified through retrospective testing of dengue-negative samples. Ongoing communication from the WHO Regional Office for the

Americas informed the international community about the virus's spread to non-endemic states like Bahia, Piauí and Espírito Santo. The Regional Office's alerts prompted enhanced surveillance in neighbouring countries, with Bolivia (Plurinational State of) confirming Oropouche through cross-border collaboration and Colombia beginning retrospective testing. In Cuba, an early rise in undifferentiated fever led to the first documented Oropouche transmission on the island. Similarly, Guyana and Barbados reported their first cases in 2024, highlighting the virus's growing presence in the northern coastal regions of South America and the Caribbean.

During the 2024 outbreak, Oropouche virus disease presented atypical severe cases, with fatal cases, neurological complications and vertical transmission, including fetal deaths

and congenital anomalies, reported by Brazil and Cuba. Throughout the 2024 Oropouche outbreak, NFPs played a crucial role in detection, verification and coordinated response efforts. Brazil's NFP promptly responded to the Regional Office request for information in January 2024, confirming the emerging outbreak. Bolivia's (Plurinational State of) NFP collaborated with Brazil's NFP to confirm Oropouche virus along shared borders (Fig. 32). The Regional Office's joint epidemiological alerts and NFP communications prompted retrospective analysis of dengue-negative samples in Colombia and testing of febrile cases in Cuba, leading to the timely detection of Oropouche virus in new areas. Effective contact between NFPs also helped identify and report imported traveler-related cases early on.

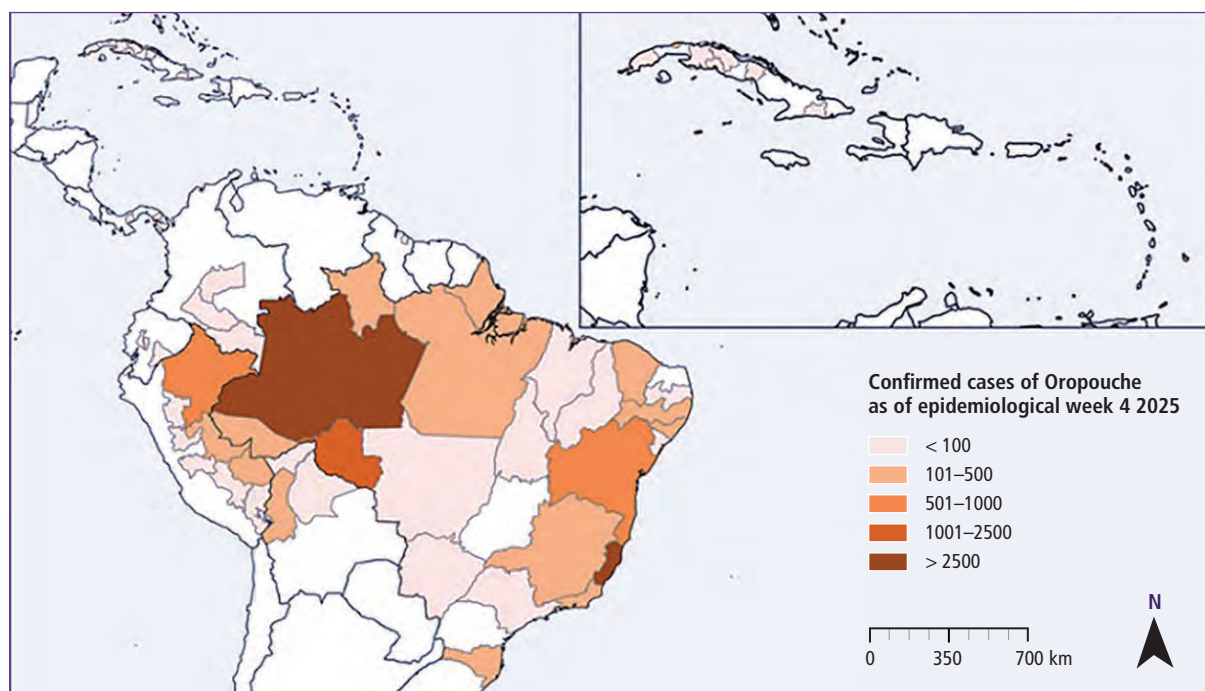
FIG. 31 Number of confirmed autochthonous cases of Oropouche by country and epidemiological week of symptom onset, WHO Region of the Americas, 1 January 2024–25 January 2025



Note: Information is up to EW 4 of 2025 for most countries; information for Brazil and Peru is up to EW 5 of 2025.

Source: Adapted and reproduced by the WHO Region of the Americas from the data reported by the respective countries.

FIG. 32 Autochthonous confirmed cases of Oropouche notified by the National IHR Focal Points in the WHO Region of the Americas, 2024–2025 (as of EW 4*)



* Information for Brazil is up to EW 5 of 2025 and information for Panama is up to EW 6 of 2025.



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Emerging trends for key diseases

In 2024, the global infectious disease landscape remained complex, with multiple ongoing acute events posing significant public health risks globally.

Cholera outbreaks persisted at an unprecedented scale, with sustained transmission in both endemic and previously unaffected areas. High case fatality rates were observed in several settings and global supplies of oral cholera vaccine remained insufficient to meet demand. The global spread of highly pathogenic avian influenza A(H5N1) raised new concerns, with increasing detections in mammals and occasional human infections. In 2024, several human cases were reported, including fatalities, underscoring the virus's potential to cause severe disease in humans and the importance of monitoring its evolution.

Vector-borne diseases showed significant expansion. Oropouche virus outbreaks were reported in South America highlighting the emergence of neglected arboviral threats associated with environmental and climate changes. Viral haemorrhagic fevers (VHFs) remained a priority for global health security and a significant public

health problem in the WHO African Region in 2024. Marburg virus disease was reported for the first time in Rwanda and Sudan virus disease re-emerged in Uganda. Both outbreaks necessitated rapid containment measures and demonstrated the continued risk of zoonotic spillover events.

After the 2022–2023 multi-country mpox PHEIC was declared over in May 2023, a new PHEIC was declared in August 2024 for the upsurge of mpox in the Democratic Republic of the Congo and a growing number of countries in the WHO African Region. Ongoing transmission has been reported in several countries, particularly affecting vulnerable and underserved populations. The risk of international spread of poliovirus continues to constitute a PHEIC, having first been declared as such in 2014, with 2024 marked by persistent challenges related to both wild poliovirus type 1 (WPV1) and cVDPV2 transmission. Multiple countries reported ongoing detection of both types, with significant outbreaks occurring in parts of Africa, the Middle East and South Asia. These trends underscore the critical importance of maintaining high immunization coverage and robust surveillance systems.

Key disease trend in the WHO African Region

The WHO African Region continues to experience a large number of public health emergencies. In 2024, a total of 141 new public health events were detected across 41 countries. Several multi-country outbreaks of infectious diseases were recorded across the Region, including cholera, mpox, dengue fever and diphtheria. In addition,

viral haemorrhagic fever outbreaks were reported in Rwanda (Marburg virus disease), the United Republic of Tanzania (Marburg virus disease) and Uganda (Sudan virus disease). Major disasters also affected the WHO African Region, notably floods, cyclones, droughts and landslides. These have exacerbated existing humanitarian

emergencies and placed already stretched health systems under considerable strain. The complex and protracted humanitarian crises in the Sahel region, the eastern Democratic Republic of Congo, South Sudan and Ethiopia (fuelled by armed conflicts, climatic shocks, disease outbreaks) have precarious health, nutrition

and socioeconomic consequences for millions of people across the Region. Moreover, in 2024, the Region was still affected by the Sudan conflict that began in April 2023, where tens of thousands of people continue to flee Sudan and to seek refuge in Chad, Ethiopia and South Sudan.

Key disease trend in the WHO Region of the Americas

During 2024, public health in the WHO Region of the Americas faced several challenges across the Region due to the occurrence of multiple events that affected the general population. These challenges included a notable increase in dengue that surpassed the year 2023, which was a record epidemic year; the emergence of Oropouche in countries and areas where cases were not previously registered, included fatal cases, cases with neurological complications and instances of vertical transmission; identification of human cases of avian influenza A(H5N1) related to outbreaks affecting birds, dairy cattle and other

mammals; and detection of first imported cases of mpox clade 1b in the Region. Additionally, in the Southern cone countries, there were reports of Western equine encephalitis in horses, with subsequent occurrence of isolated human cases. Yellow fever outbreaks were also identified, including areas outside the Amazon region. The confirmation of circulating vaccine-derived poliovirus type 3 (cVDPV3) detected in wastewater samples. Identification of measles clusters and cases in countries and territories in the Region. Multiple countries have experienced concurrent outbreaks of two or more arboviruses.

Key disease trend in the WHO Eastern Mediterranean Region

In 2024, the WHO Eastern Mediterranean Region faced many challenges, including vector-borne diseases, especially dengue outbreaks in several countries [Afghanistan, Djibouti, Iran (Islamic Republic of), Oman, Pakistan, Qatar, Saudi Arabia, Somalia, Sudan, the United Arab Emirates and Yemen]. The climate change, including heavy rains and floods in many countries, led to an increase in dengue fever cases, including in countries such as Iran (Islamic Republic of), Qatar and the United Arab Emirates where autochthonous cases were detected for the first time ever in the end of 2023 and beginning of 2024. This environmental phenomenon has also resulted in an increase in the trend of water-borne diseases, especially cholera, in many countries, making the Region the major contributor to the global burden of the disease.

Other vaccine-preventable disease outbreaks were also reported across some of the Region's countries, including cVDPV-2, measles and diphtheria, underscoring the need to fill the gap in vaccination coverage in those countries. The armed conflict in West Bank and Gaza Strip led to disruption of the health services, including immunization activities, which led to an outbreak of cVDPV-2, including a human case. Additionally, increases in measles cases were observed in countries such as Afghanistan, Djibouti, Iran (Islamic Republic of), Iraq, Lebanon, Libya, Morocco, Pakistan, Somalia, Sudan, Syrian Arab Republic and Yemen. Most of these countries have fragile health systems due to either armed conflicts or political instability.

Key disease trend in the WHO European Region

In 2024, the WHO European Region reported 63 public health events across 24 countries, territories, and areas. Most events were of an infectious nature, followed by natural disasters. There have been reports of emerging and re-emerging diseases, with the most common infectious events being associated with zoonoses, vector-borne diseases and vaccine-preventable diseases (e.g. measles and environmental detections of poliovirus).

Highly pathogenic avian influenza (HPAI) H5 was widely detected among wild birds, poultry and mammals, as continued from the ongoing outbreak that began in previous years. Several events were associated with zoonoses, including various reports of mpox virus (MPXV). An increased incidence of imported cases of clade I MPXV were reported among several regional countries with epidemiological links to countries within the African region currently experiencing an active mpox outbreak. There were several detections of polio in environmental samples, specifically cVDPV2—these were observed across various countries, including Finland, Germany, Poland, Spain and the United Kingdom.

In August 2024, the French overseas territory of La Réunion reported the first autochthonous cases of chikungunya, with over 100 cases reported by the end of the year — this was the first large outbreak of chikungunya since 2005–2006. Vector-borne diseases were reported with an increased frequency during increased vector activity season, including an expanded geography of autochthonous cases of dengue in the southern part of the Region, including in France, Italy and Spain, as well as increased reports of West Nile virus.

The Region also faced a number of natural hazards, including extreme weather events. Floods were reported in a number of countries and areas in the Region, including Spain, where extreme floods resulted in over 200 deaths. Additionally, the French overseas department and region of Mayotte experienced a significant impact from Cyclone Chido, a severe natural disaster that caused extensive damage to residential areas and resulted in over 170 fatalities. Furthermore, with climate change, there has been an increasing number of extreme weather events and natural hazards affecting the Region.

Key disease trend in the WHO South-East Asia Region

Cholera and acute watery diarrhoea (AWD) continue to affect the WHO South-East Asia Region, with Bangladesh, India, Myanmar and Nepal reporting the cases. In particular, Myanmar experienced a major outbreak, starting in Yangon and spreading to at least nine regions/states, for which reactive oral cholera vaccine campaign was implemented in selected areas.

Vector-borne diseases continue to pose a high burden. Indonesia experienced a large dengue outbreak, while Timor-Leste recorded a chikungunya outbreak for the first time. India and Thailand recorded outbreaks of Zika virus disease,

while India experienced an outbreak of Chandipura virus infection.

Zoonotic threats continue. Cases of Nipah virus infection were reported in Bangladesh and India, with a high case fatality rate (all seven cases were fatal). Human infections of avian influenza (A(H9N2) and A(H5N1)) occurred in India. Timor-Leste, a previously rabies-free country, reported the first human case of rabies. Outbreaks of leptospirosis affected India, Indonesia, Sri Lanka and Thailand. An increase of Kyasanur Forest disease cases was recorded in India.

Travel-related mpox cases, associated with clade 1b, were documented in Thailand and India. Detection of polio virus was reported in Indonesia and early 2024, including circulating Vaccine Derived poliovirus type 2 and poliovirus type 1. Sri Lanka has experienced an outbreak of measles.

The Region is also prone to major impact from natural disasters, with several cyclones, floods and volcanic eruptions recorded in various countries throughout 2024, one of which was cyclone Yagi, which triggered severe floods in Myanmar and Thailand.

Key disease trend in the WHO Western Pacific Region

In 2024, a total of 72 public health events were reported across 37 countries and areas in the WHO Western Pacific Region. The Region continued to experience sporadic human infections with avian influenza A, including both high- and low-pathogenic avian influenza. China reported human infections with H5N1, H5N6, H9N2, H10N3 and H10N5, with H10N5 reported in humans for the first time globally. Australia reported its first human infection with H5N1, while Cambodia and Viet Nam observed a re-emergence of H5N1 human cases. Some of these cases resulted in severe illnesses and fatalities. A novel reassortant H5N1 clade 2.3.2.1e (previously classified as 2.3.2.1c) was detected in human cases reported in Cambodia and Viet Nam. Additionally, outbreaks of vaccine-preventable diseases, such as pertussis, measles and diphtheria, were reported across multiple Member States, including Pacific Island countries and areas. A dengue outbreak was also officially declared in Fiji and Samoa. Furthermore, two countries reported cases of diseases for the first time: Q fever in the Philippines and an imported case of Lassa fever in China.

The Region additionally faced various natural hazards in 2024. During the northern hemisphere winter, Mongolia experienced Dzud – an extreme winter event characterised by temperatures dropping to – 30°C or lower, strong blizzards, heavy snowfall and ice, followed by flash floods in spring due to snow and ice melting. The 2024 Dzud was the most severe in the past 50 years, significantly impacting rural populations due to massive loss of livestock, infrastructure damage, disruptions to primary healthcare, and increased health risks.

Typhoons and cyclones also posed major challenges. Among them, super typhoon Yagi affected multiple countries in the Region, including China, the Philippines and Viet Nam, resulting in casualties, large-scale evacuations and infrastructure damage. Lastly, in mid-December 2024, a 7.3 magnitude earthquake struck Vanuatu, affecting over 80 000 people, including 14 deaths and over 200 injuries.

Conclusion

This report reaffirms the essential role of PHI activities in enabling the detection, verification, risk assessment and timely dissemination of information on acute public health events globally under the International Health Regulations. It highlights the evolving landscape of global health threats, and the growing need for coordinated, timely and evidence-based action.

In 2024, a total of 2805 signals of potential public health concern were detected, representing a 91% increase compared to 2023. This sharp rise reflects both the strengthening of WHO's PHI systems and the increasing number of acute public health threats globally. At the same time, the proportion of signals verified within 24 to 48 hours under the IHR continued to increase, confirming ongoing improvements in early risk assessment and verification workflows, and highlighting the essential role of States Parties and of NFPs.

In the context of climate change, these hazards are becoming more frequent, more intense and more interconnected, creating complex and cascading risks for communities and systems. The detection of those signals and the activation of the early warning systems are essential for reducing the growing risks posed by climate-related hazards.

Events due to avian or animal origin influenza emerged as the most frequent cause of acute public health events in 2024, reflecting the growing global concern surrounding zoonotic spillovers. The public health relevance of these events is reflected in the information products produced: 39 EIS bulletins focused on avian influenza, accounting for 43%

of all bulletins issued. Similarly, DON reports were predominantly issued for events related to zoonotic influenza, while the overall number of DON reports declined due to the operational shift of cholera and mpox reporting to more in-depth situation reports.

While the total number of RRAs declined in 2024, there was a substantial increase in the number of global RRAs, reflecting the growing number of public health events with cross-border or multi-country relevance. This trend underscores the global nature of today's health threats and the critical need for coordinated international action. Among these, cholera RRAs showed an upward trend, indicating a shift in priorities and a resurgence of high-burden communicable diseases in vulnerable settings.

Timeliness remains a cornerstone of effective public health intelligence. In 2024, the median time to detect and respond to events remained within the benchmark set by the 7–17 indicator, while notification time to WHO exceeded the 1-day adapted target, pointing to an area requiring further attention and improvement. Regarding information products, DON reports consistently met expected timelines, while RRA and EIS bulletins showed delays. Enhancing collaboration with NFPs, ensuring sufficient human resource capacity and improving internal workflows will be key to ensuring WHO meets its IHR obligations for timely communication with States Parties.

The diversity of health hazards — from zoonotic outbreaks, disasters, armed conflicts to product-related events — reaffirms the need for a

comprehensive, multi-hazard approach. Collaboration with platforms such as GLEWS+, and the integration of regional analyses and risk communication efforts, continue to strengthen WHO's capacity to monitor and respond to emerging threats at the human-animal-ecosystem interface.

In a context marked by increasing geopolitical instability, resurgence of epidemic-prone diseases, climate-related shocks and global financial constraints, PHI remains a vital pillar of WHO's emergency preparedness and response mandate. Sustained investment, improved timeliness, and enhanced collaboration across

sectors and regions are indispensable to protect health, prevent crises, and guide early and effective decision-making.

Above all, public health intelligence continues to be a cornerstone of the implementation of the International Health Regulations. Through timely detection, verification and communication of health threats, PHI strengthens the global health architecture, and supports WHO Member States in preventing and responding to events that may constitute a public health emergency of international concern.

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NOTE

For all the figures/tables in the annexes, the legends for the WHO regions are:

- AFR:** WHO African Region
- AMR:** WHO Region of the Americas
- EMR:** WHO Eastern Mediterranean Region
- EUR:** WHO European Region
- SEAR:** WHO South-East Asia Region
- WPR:** WHO Western Pacific Region

Annex 1

States Parties and areas by WHO region

WHO African Region

The WHO African Region consists of the following 47 States Parties: Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau,

Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, South Africa, South Sudan, Togo, Uganda, United Republic of Tanzania, Zambia, Zimbabwe.

WHO Region of the Americas

The WHO Region of the Americas consists of the following 35 States Parties: Antigua and Barbuda, Argentina, the Bahamas, Barbados, Belize, Bolivia (Plurinational State of), Brazil, Canada, Chile, Colombia, Costa Rica, Cuba, Dominica, the Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana,

Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, the United States of America, Uruguay, Venezuela (Bolivarian Republic of).

WHO Eastern Mediterranean Region

The WHO Eastern Mediterranean Region consists of the following 21 State Parties and one territory: Afghanistan, Bahrain, Djibouti, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman,

occupied Palestinian territory, including east Jerusalem, Pakistan, Qatar, Saudi Arabia, Somalia, Sudan, Syrian Arab Republic, Tunisia, United Arab Emirates, Yemen.

WHO European Region

The WHO European Region consists of the following 55 States Parties: Albania, Andorra, Armenia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Holy See, Hungary, Iceland, Ireland, Israel, Italy, Kazakhstan, Kyrgyzstan, Latvia, Liechtenstein, Lithuania, Luxembourg,

Malta, Monaco, Montenegro, Netherlands (Kingdom of the), North Macedonia, Norway, Poland, Portugal, Republic of Moldova, Romania, Russian Federation, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Tajikistan, Türkiye, Turkmenistan, Ukraine, United Kingdom of Great Britain and Northern Ireland, Uzbekistan.*

WHO South-East Asia Region

The WHO South-East Asia Region consists of the following 11 States Parties: Bangladesh, Bhutan, Democratic People's Republic of

Korea, India, Indonesia, Maldives, Myanmar, Nepal, Sri Lanka, Thailand, Timor-Leste.

WHO Western Pacific Region

The WHO Western Pacific Region consists of the following 37 States Parties and areas: American Samoa (*United States of America*), Australia, Brunei Darussalam, Cambodia, China, Hong Kong SAR (*China*), Macao SAR (*China*), Cook Islands, Fiji, French Polynesia (*France*), Guam (*United States of America*),

Japan, Kiribati, Lao People's Democratic Republic, Malaysia, Marshall Islands, Micronesia (Federated States of), Mongolia, Nauru, New Zealand, Niue, Palau, Papua New Guinea, Philippines, Republic of Korea, Samoa, Singapore, Solomon Islands, Tonga, Tuvalu, Vanuatu, Viet Nam.

* The Holy See and Liechtenstein are also States Parties to the IHR (2005), but they are not Member States of WHO.

Annex 2: SUPPLEMENTARY FIGURES

FIG. A Acute public health events for the ten most frequently reported diseases, conditions or hazards of an infectious nature, by WHO region, 2020–2024

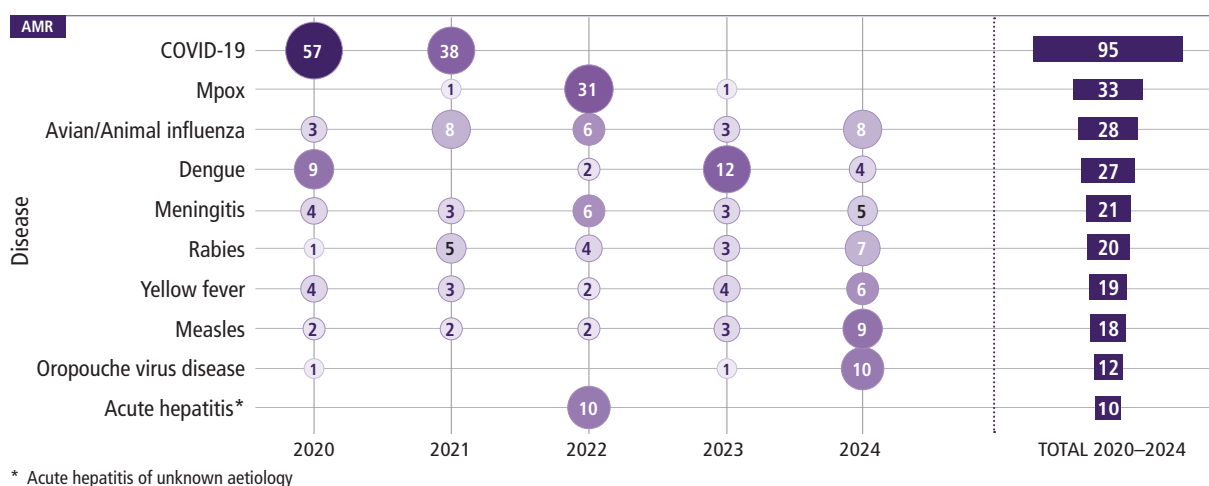
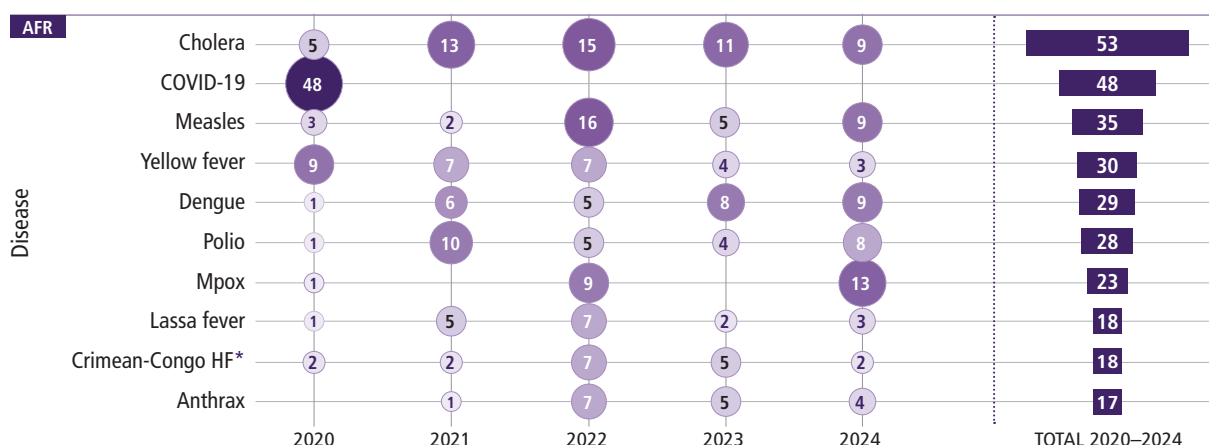


FIG. A Acute public health events for the ten most frequently reported diseases, conditions or hazards of an infectious nature, by WHO region, 2020–2024 (*continued*)



Source: WHO.

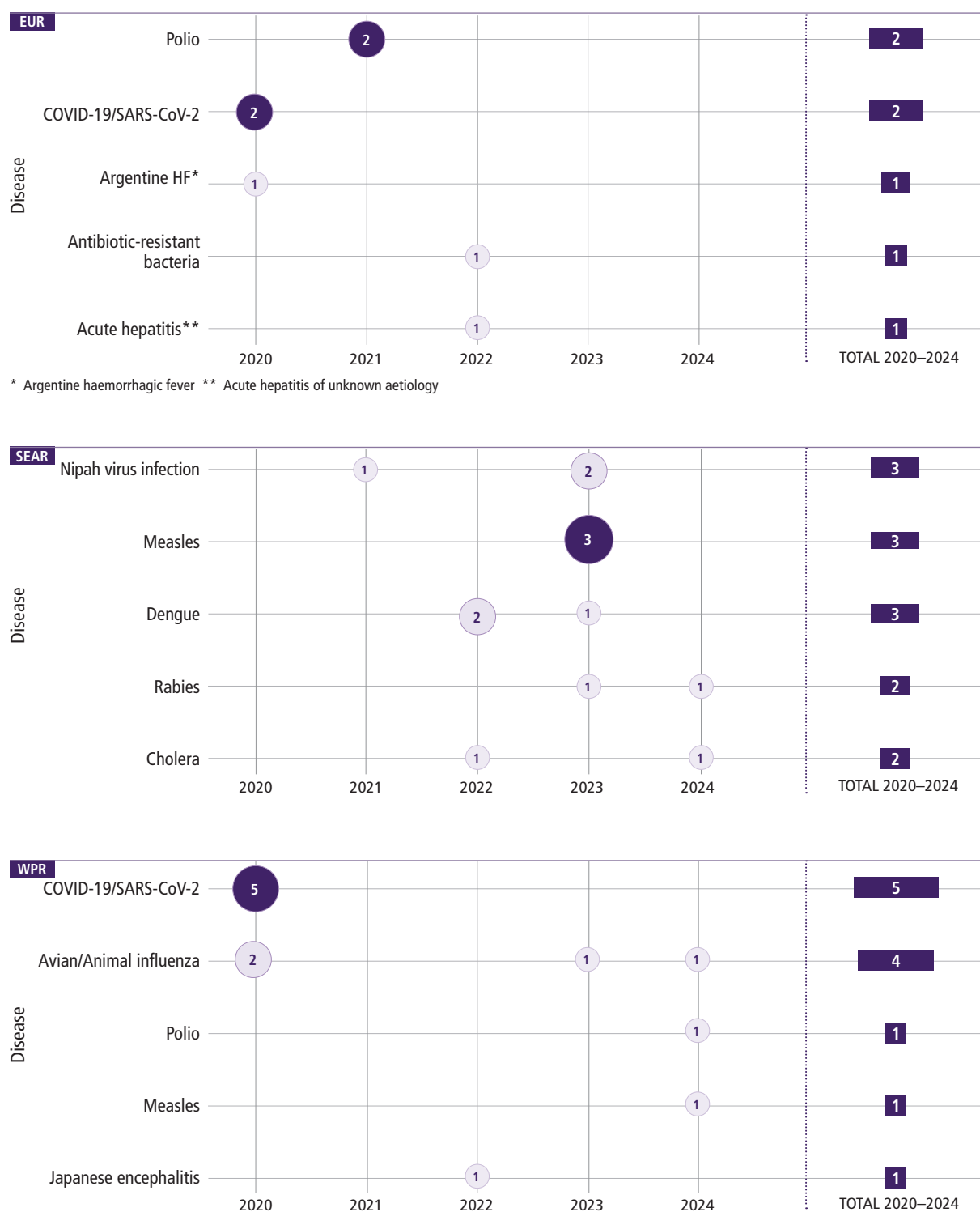
WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

FIG. B Rapid Risk Assessments by WHO region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024



FIG. B Rapid Risk Assessments by WHO region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024 (*continued*)



Source: WHO.

WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

FIG. C Event Information Site (EIS) bulletins by region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024

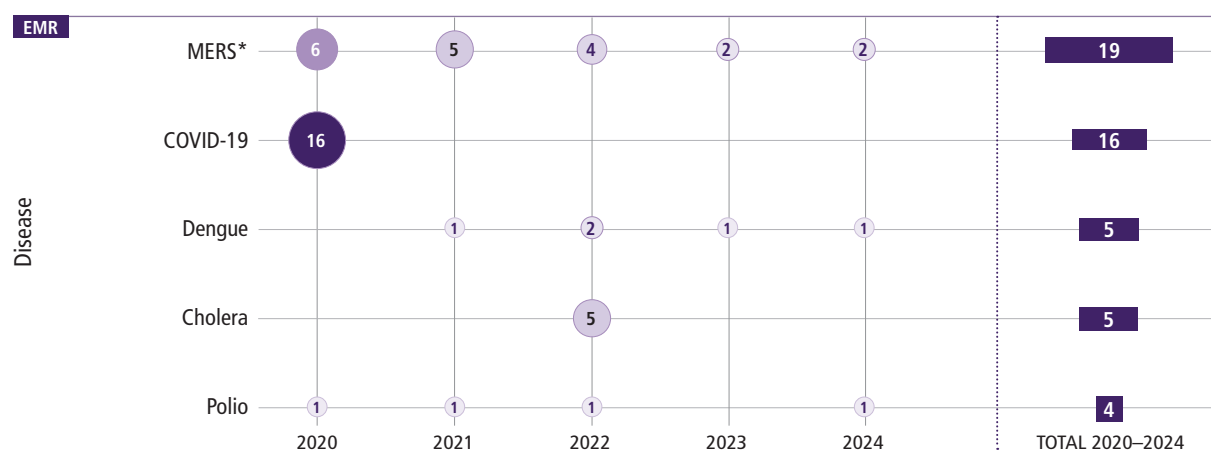
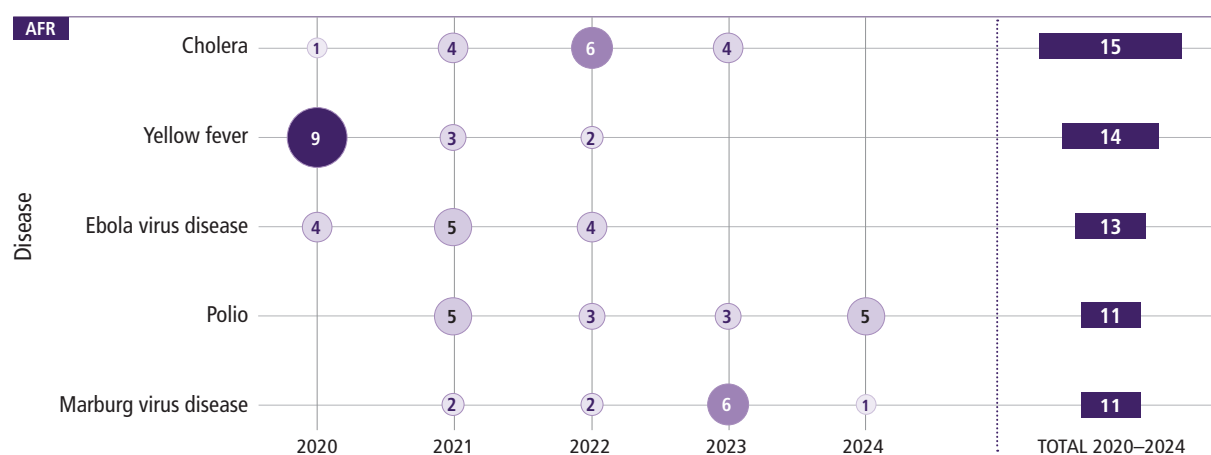


FIG. C Event Information Site (EIS) bulletins by region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024 (*continued*)

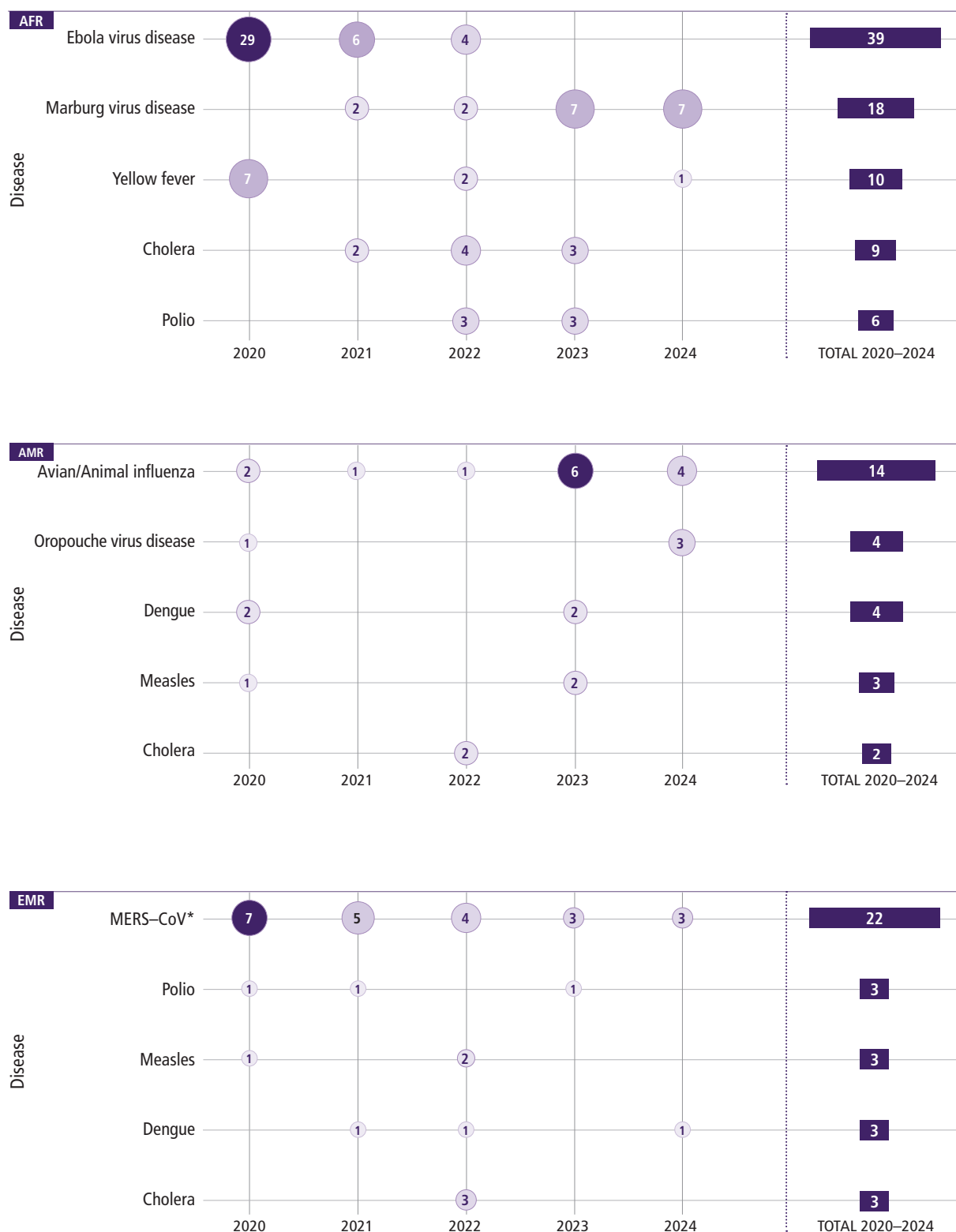


Source: WHO.

WHO regions

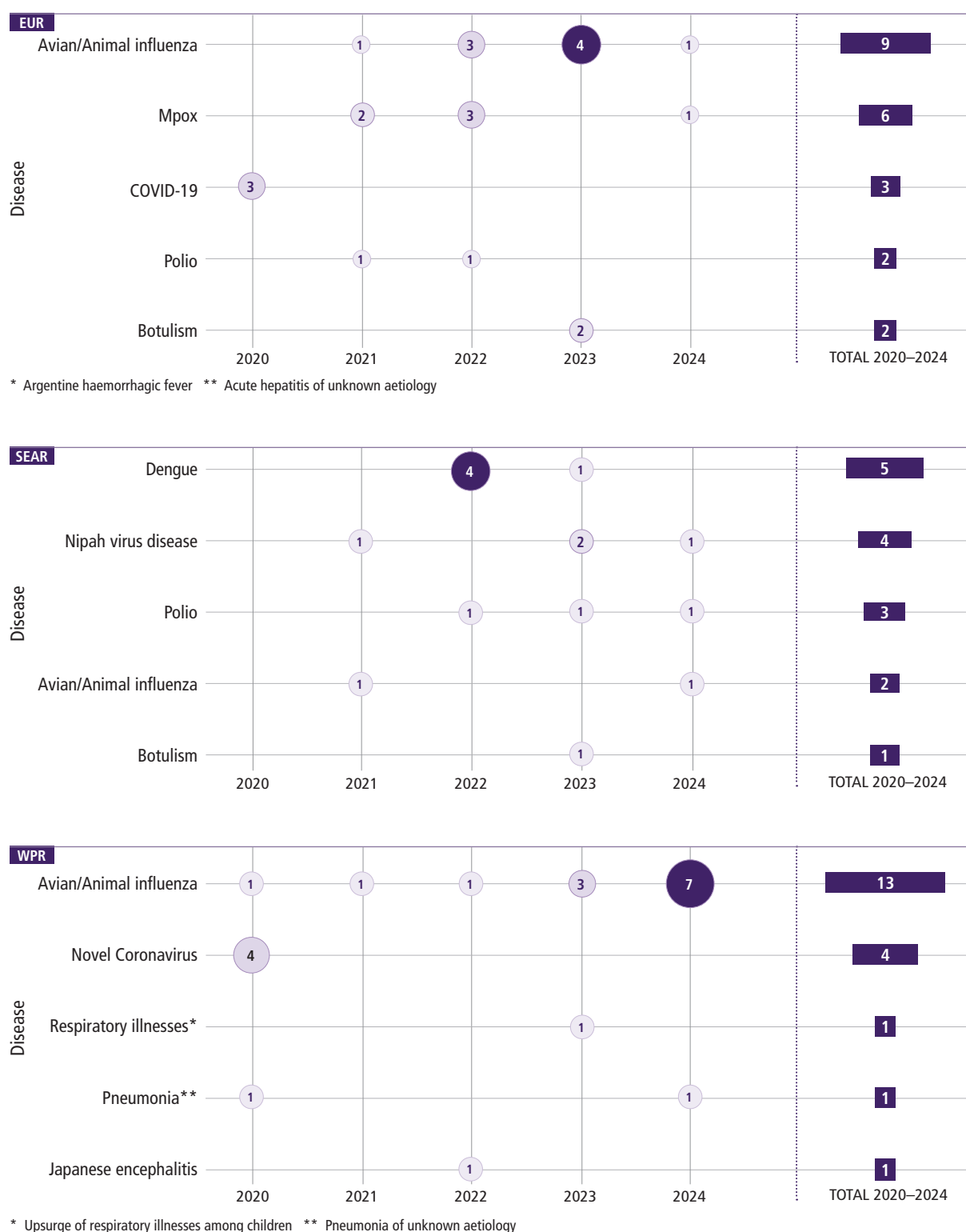
AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

FIG. D Disease Outbreak News (DON) reports by region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024



* MERS, Middle East respiratory syndrome coronavirus

FIG. D Disease Outbreak News (DON) reports by region for the five most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and aggregated across the five-year period, 2020–2024 (*continued*)



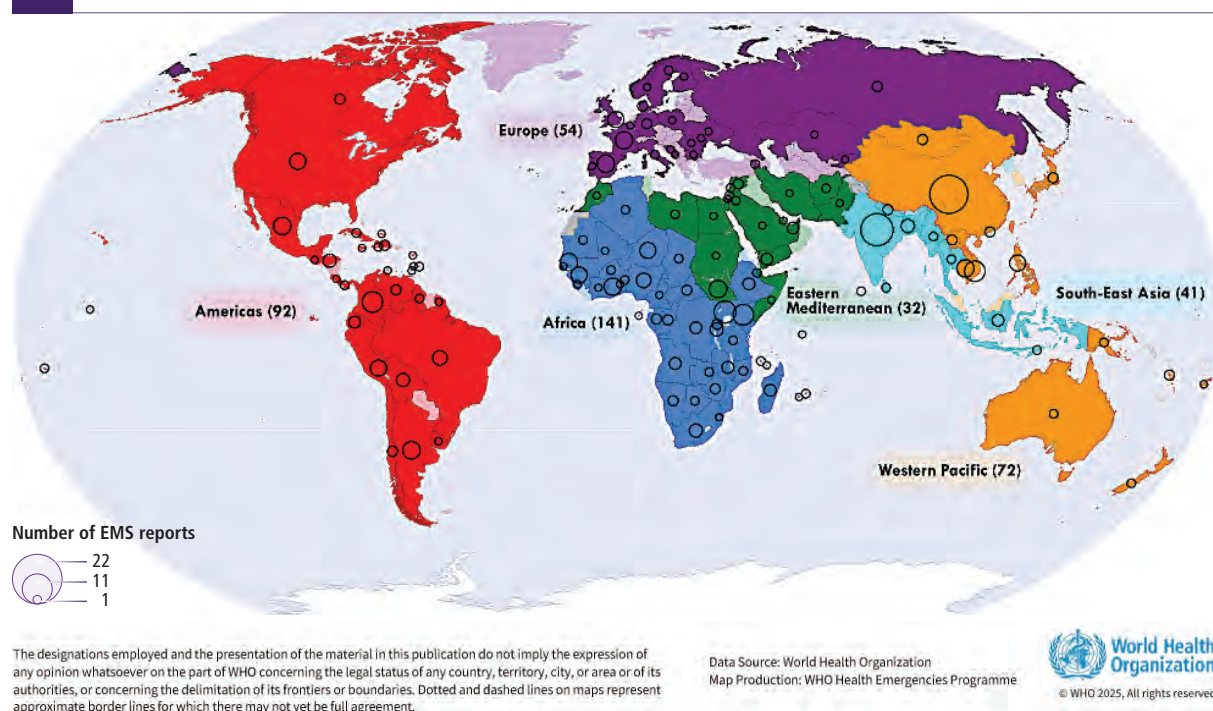
Source: WHO.

WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

FIG. E MAPS of: acute public health events (E1), EIS bulletins (E2), RRA reports (E3) and DON reports (E4), 2024

» E1 ACUTE PUBLIC HEALTH EVENTS, 2024



» E2 EVENT INFORMATION SITE (EIS) bulletins, 2024

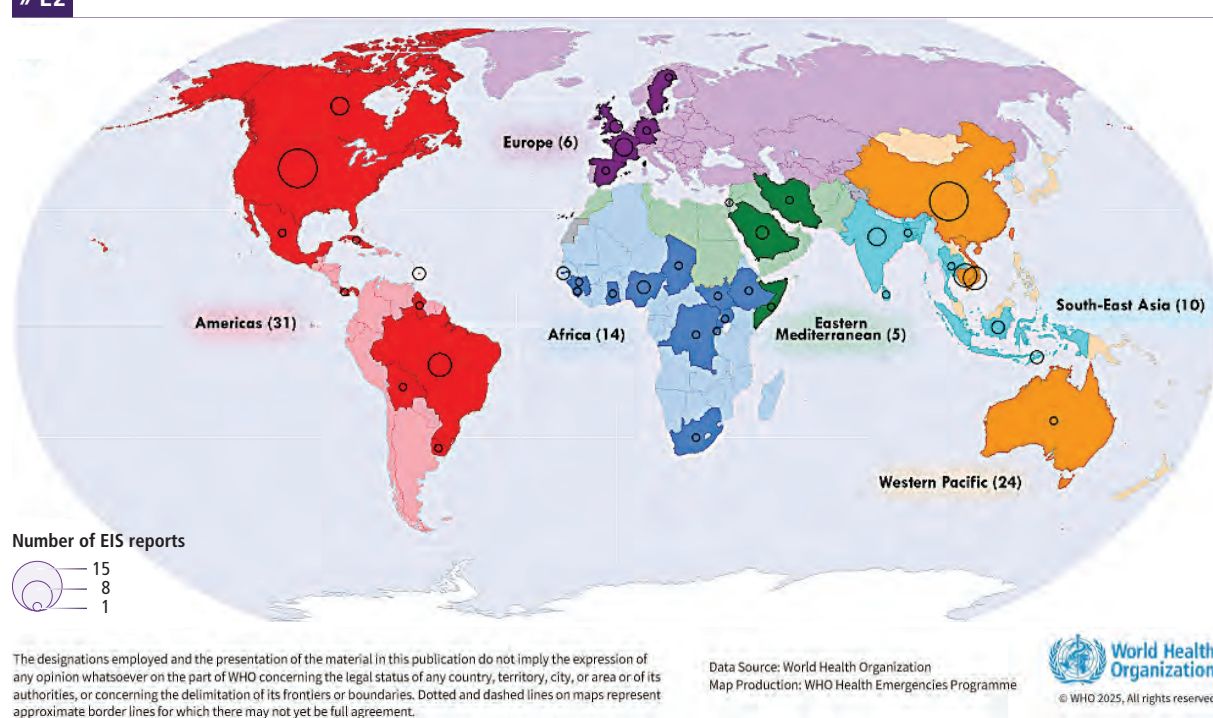
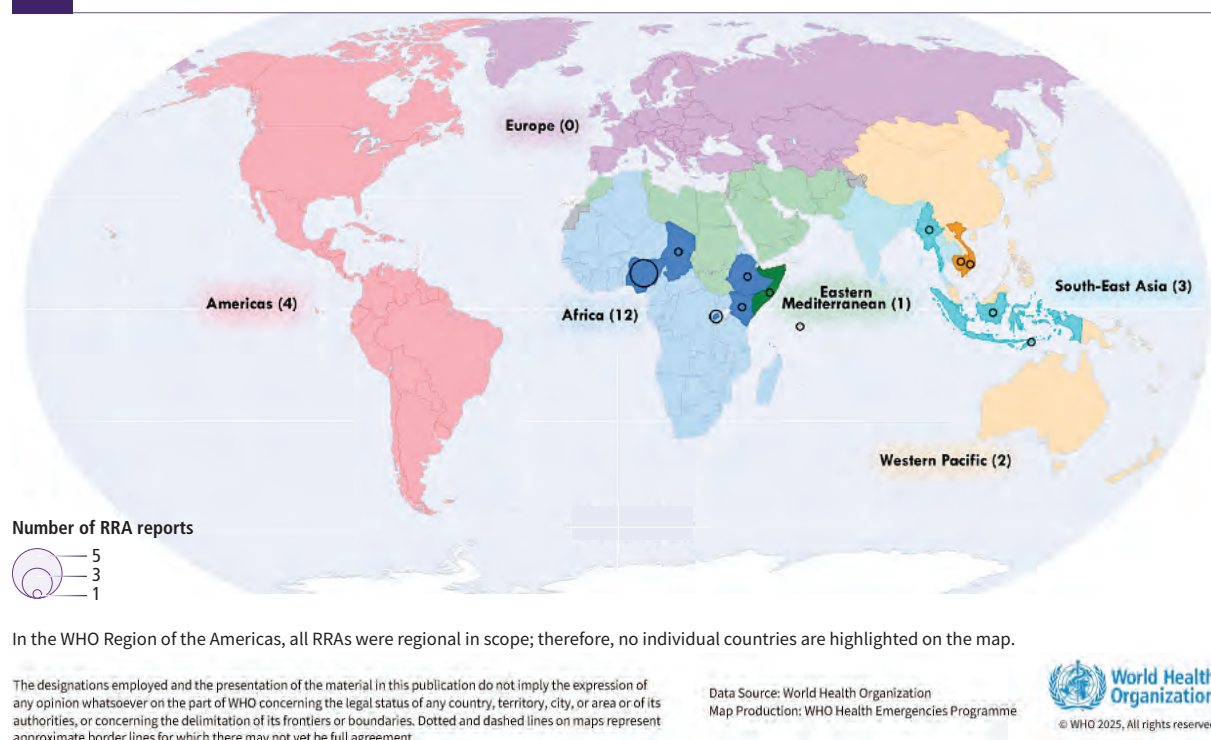


FIG. E MAPS of: acute public health events (E1), EIS bulletins (E2), RRA reports (E3) and DON reports (E4), 2024
(continued)

» **E3** **RAPID RISK ASSESSMENT (RRA) reports, 2024**



» **E4** **DISEASE OUTBREAK NEWS (DON) reports, 2024**

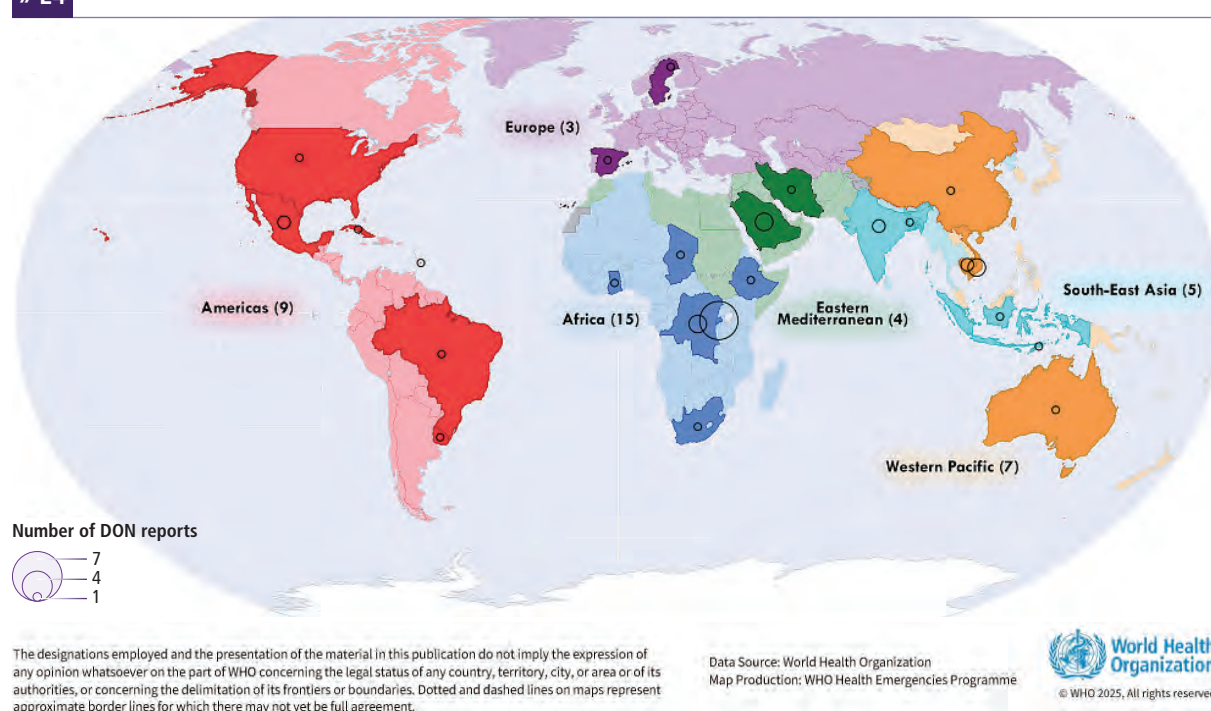
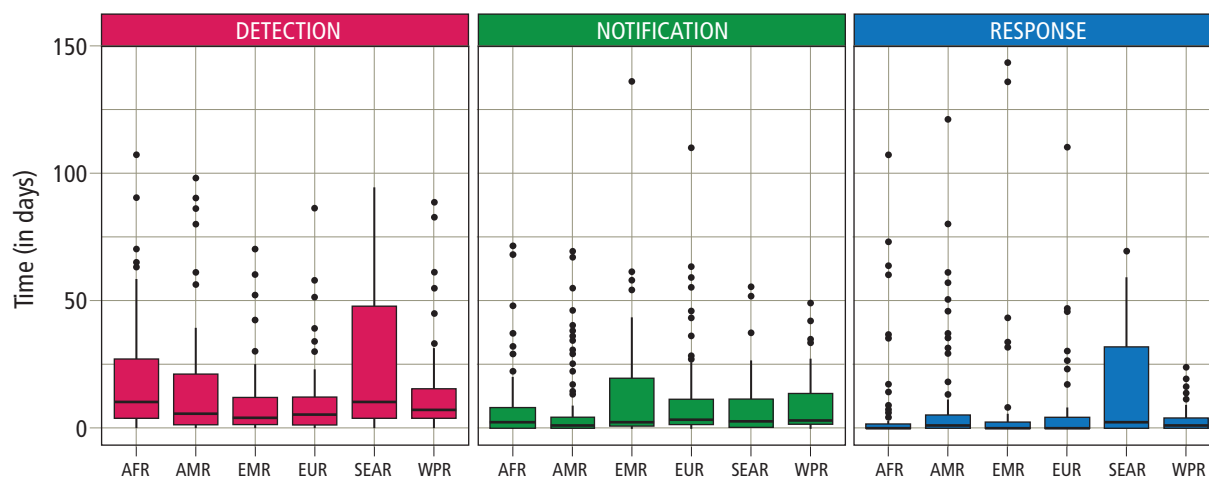


FIG. F Timeliness of events published in the Event Information Site (EIS), in days, for detection, notification and response, by WHO region, 2020–2024

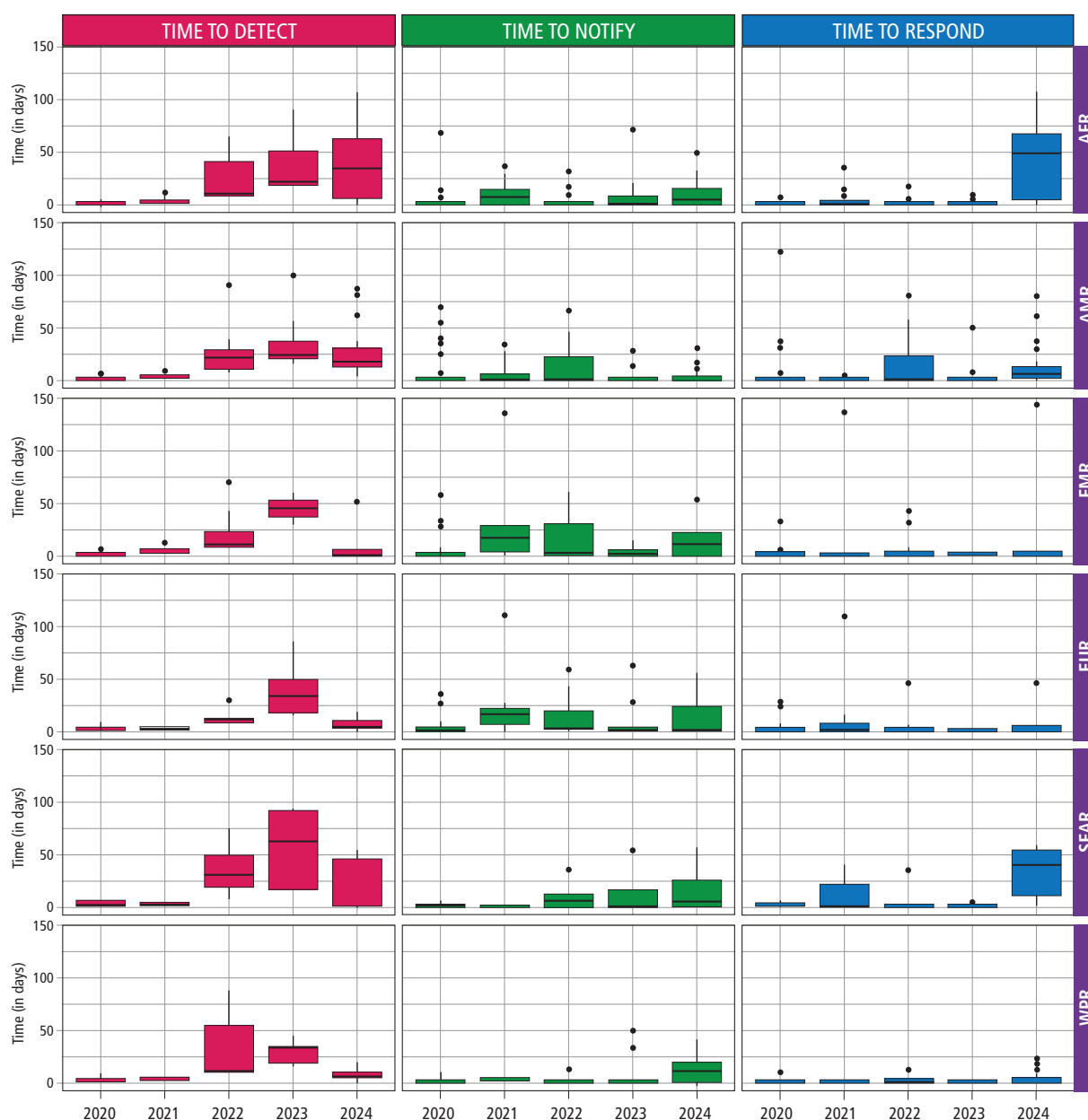


WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

Source: WHO.

FIG. G Timeliness of events published in the Event Information Site (EIS), in days, for detection, notification and response, by WHO region, for each year of the past five years, 2020–2024



WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

Source: WHO.

Annex 3

Regional information dissemination for acute public health events

There are numerous regional information products that provide essential and timely information on acute public health events within WHO regions. Annex 3 provides a non-exhaustive list of these by WHO region.

WHO African Region

- ▶ Weekly bulletin: Weekly bulletins on outbreaks and other emergencies, WHO Regional Office for Africa
- ▶ Event tracker: WHO African Region Dashboard
- ▶ Dengue sitreps: Dengue, WHO Regional Office for Africa
- ▶ Cholera sitreps: Cholera, WHO Regional Office for Africa
- ▶ Mpox in Africa
- ▶ Meningitis Dashboard: Meningitis, WHO Regional Office for Africa

WHO Region of the Americas

- ▶ Epidemiological Alerts and Updates
- ▶ Public Rapid Risk assessments
- ▶ Briefing notes and laboratory guidelines
- ▶ Daily Natural Hazards Monitoring
- ▶ Daily Report of Events
- ▶ Select Signals/Events daily digest (internal)
- ▶ Near-real-time natural hazards in the Americas: Population and emergency hospitals exposure (Dashboard)
- ▶ Situation Reports
- ▶ Public Health Situation Analysis
- ▶ Public health signals and events being monitoring dashboards
- ▶ Mpox cases in the Americas Region (Dashboard)
- ▶ Cholera resurgence in Hispaniola (Dashboard)

- ▶ Western equine encephalitis (Dashboard)
- ▶ Influenza A(H5N1) (Dashboard)
- ▶ ICC Men's T20 World Cup 2024 (Dashboard)
- ▶ Yellow Fever Vaccination Recommendations in the Americas (Dashboard)
- ▶ Meningococcal Disease Surveillance (SIREVA) (Dashboard)

WHO Eastern Mediterranean Region

- ▶ Weekly sitreps for Sudan escalation of violence and humanitarian crisis
- ▶ Weekly sitreps for Humanitarian impact of the current situation in occupied Palestinian territory, including east Jerusalem
- ▶ Morocco Earthquake: information for briefing notes
- ▶ Daily Eastern Mediterranean Regional Surveillance Report
- ▶ WHO/EMRO Weekly Summary of Events
- ▶ EMRO health emergencies quarterly bulletin
- ▶ Morocco earthquake sitreps
- ▶ Türkiye and Syrian Arab Republic earthquake sitreps
- ▶ Health emergencies biweekly bulletin
- ▶ Daily EBS Hajj Report

WHO European Region

- ▶ Quarterly Operational Update (for all graded emergencies)
- ▶ Internal Response Dashboard + OSL (for all graded emergencies)
- ▶ Quarterly Ukraine Bulletin (including refugee-hosting countries)
- ▶ Monthly donor report (done by the WHO Office in Gaziantep) for North West Syria
- ▶ COVID-19 transition plan. ECDC Daily and Weekly Communicable Diseases Threat Reports
- ▶ The European Respiratory Virus Surveillance Summary (ERVISS)
- ▶ EURO Daily List of Signals & Events (internal)
- ▶ Acute Events Daily Digest (internal)
- ▶ Mpox epipack (monthly, internal)
- ▶ Covid-19 epipack (monthly)
- ▶ Cyclone Chido PHSA
- ▶ Mpox Surveillance bulletin (joint ECDC–WHO/EUR)
- ▶ Medevac UKR bulletin (monthly)
- ▶ UEFA dashboard

WHO South-East Asia Region

- ▶ WHO South-East Asia Region Epidemiological Bulletin
- ▶ Myanmar: Acute Watery Diarrhea /Cholera Outbreak External Situation Report
- ▶ Public Health Situation Analysis, Myanmar, December 2024
- ▶ Technical Brief: Enhancing readiness for a Nipah virus event in countries not reporting a Nipah virus event
- ▶ Technical Brief (interim) and Priority Actions: Enhancing Readiness for mpox in WHO South-East Asia Region
- ▶ WHO South-East Asia Region Weekly Situation Updates (internal)
- ▶ Daily signal/event update (internal)

WHO Western Pacific Region

- ▶ Avian Influenza weekly updates
- ▶ Bi-weekly dengue situation updates
- ▶ Bi-weekly seasonal influenza updates
- ▶ Bi-weekly COVID-19 situation updates
- ▶ Daily summary of signals and events (internal)
- ▶ Weekly Public Health Intelligence Reports (internal)
- ▶ Two News releases on Dzud (severe winter conditions) emergency in Mongolia
- ▶ Two feature stories on avian influenza in Cambodia
- ▶ A feature story on measles response in Mongolia
- ▶ A feature story on earthquake in Vanuatu
- ▶ A feature story on dengue outbreaks in the Western Pacific Region
- ▶ A feature story on tropical storms, typhoons and cyclones in the Western Pacific

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